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(54) PATH DROPOUT FOR NATURAL LANGUAGE PROCESSING

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G06F 40/279 (2020.01)
(Continued)

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CPC ***G10L 15/063*** (2013.01); ***G06F 40/279*** (2020.01); ***G06F 40/30*** (2020.01); ***G10L 15/1815*** (2013.01); ***G10L 15/22*** (2013.01)

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See application file for complete search history.

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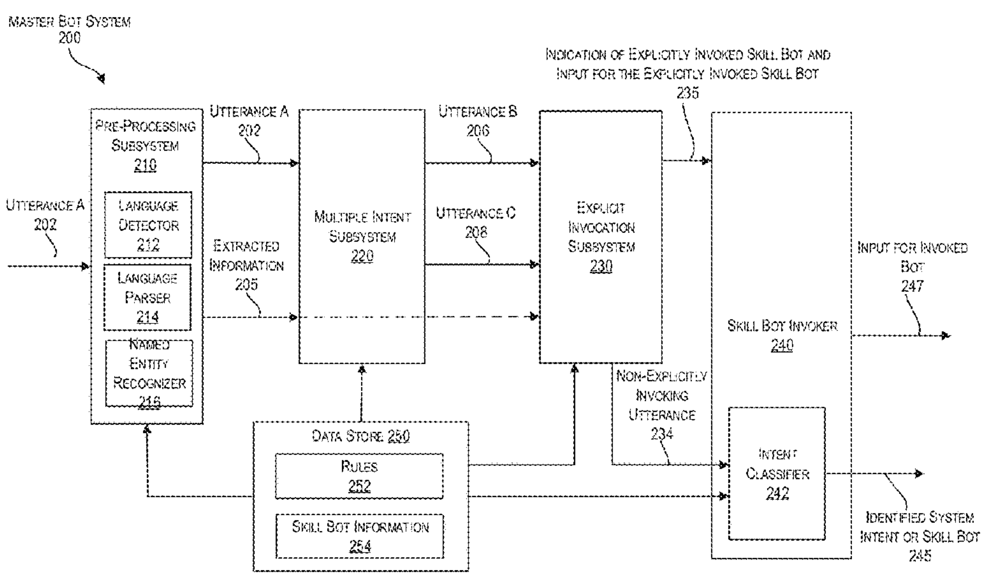
(74) *Attorney, Agent, or Firm* — Kilpatrick Townsend & Stockton LLP

(57) ABSTRACT

Techniques are provided for improved training of a machine-learning model that includes multiple layers and is configured to process textual language input. The machine-learning model includes one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second (residual) connection for providing the input to a normalization layer, bypassing the multi-head self-attention network. During training, the second connection is dropped out according to a dropout parameter. Additionally, or alternatively, an attention weight matrix is used for dropout by blocking diagonal entries in the attention weight matrix. As a result, the machine-learning model increasingly focuses on contextual information, which provides more accurate language processing results.

20 Claims, 10 Drawing Sheets



(51) **Int. Cl.**

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G10L 15/06	(2013.01)
G10L 15/18	(2013.01)

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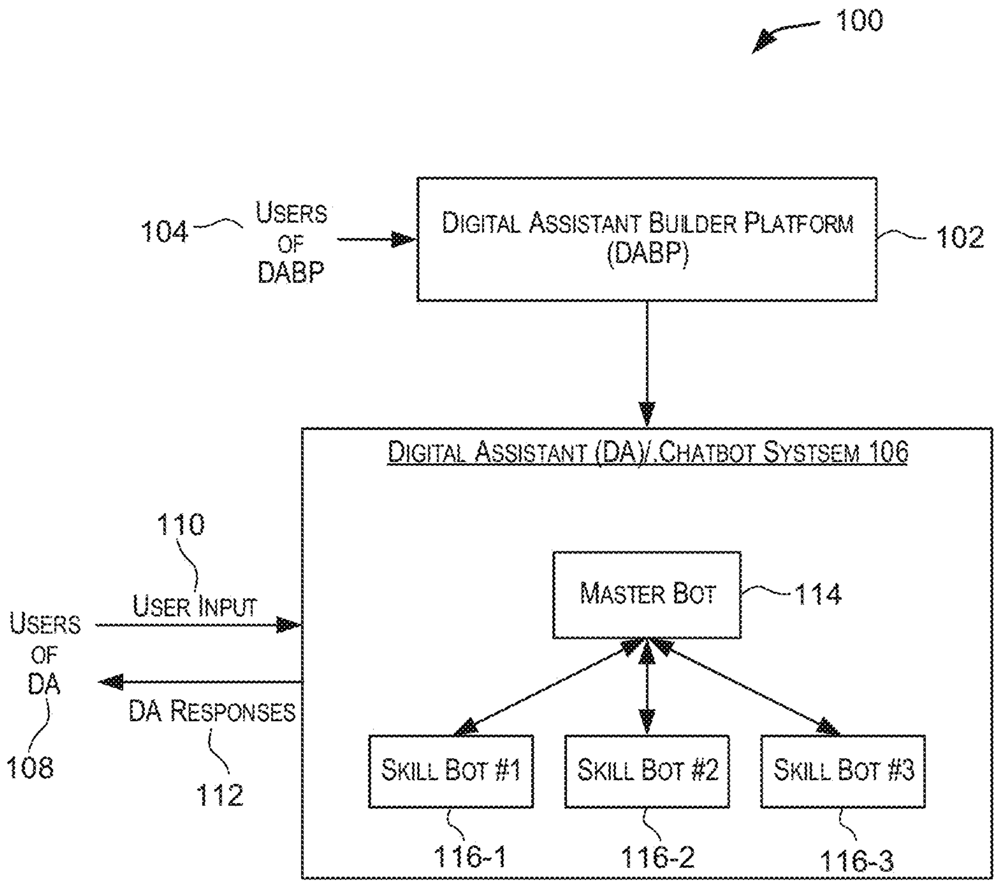

100
104
USERS OF DABP
DIGITAL ASSISTANT BUILDER PLATFORM (DABP)
102
DIGITAL ASSISTANT (DA)/ CHATBOT SYSTSEM 106
110
USER INPUT
USERS OF DA
108
DA RESPONSES
112
MASTER BOT
114
SKILL BOT #1
SKILL BOT #2
SKILL BOT #3
116-1
116-2
116-3

FIG. 1

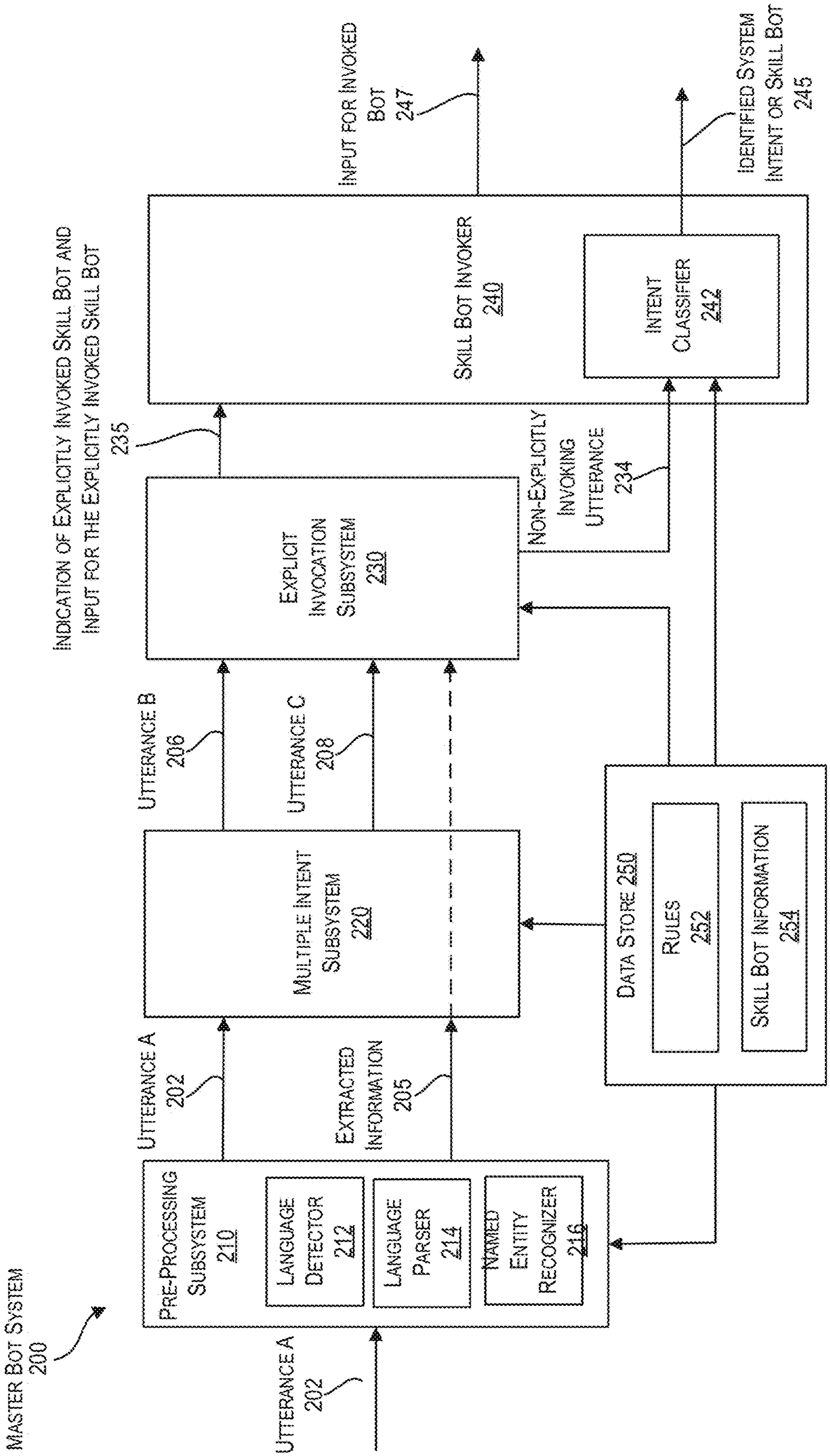
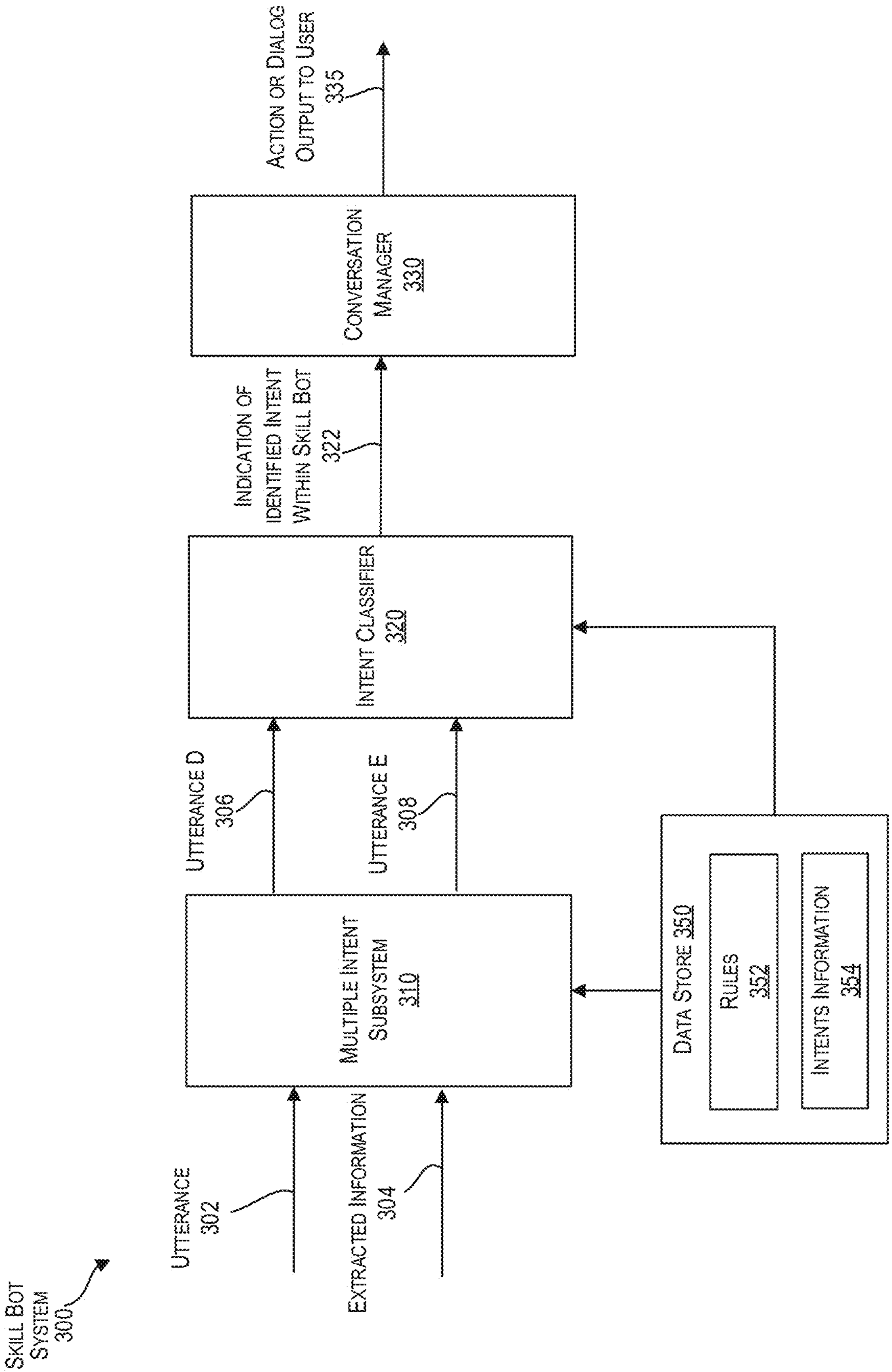

Master Bot System 200
Utterance A 202
Pre-Processing Subsystem 210
Language Detector 212
Language Parser 214
Named Entity Recognizer 216
Utterance A 202
Extracted Information 205
Multiple Intent Subsystem 220
Data Store 250
Rules 252
Skill Bot Information 254
Utterance B 206
Utterance C 208
Explicit Invocation Subsystem 230
Non-Explicitly Invoking Utterance 234
Indication of Explicitly Invoked Skill Bot and Input for the Explicitly Invoked Skill Bot 235
Skill Bot Invoker 240
Intent Classifier 242
Input for Invoked Bot 247
Identified System Intent or Skill Bot 245

FIG. 2



SKILL BOT SYSTEM 300
UTTERANCE 302
EXTRACTED INFORMATION 304
MULTIPLE INTENT SUBSYSTEM 310
UTTERANCE D 306
UTTERANCE E 308
DATA STORE 350
RULES 352
INTENTS INFORMATION 354
INTENT CLASSIFIER 320
INDICATION OF IDENTIFIED INTENT WITHIN SKILL BOT 322
CONVERSATION MANAGER 330
ACTION OR DIALOG OUTPUT TO USER 335

FIG. 3

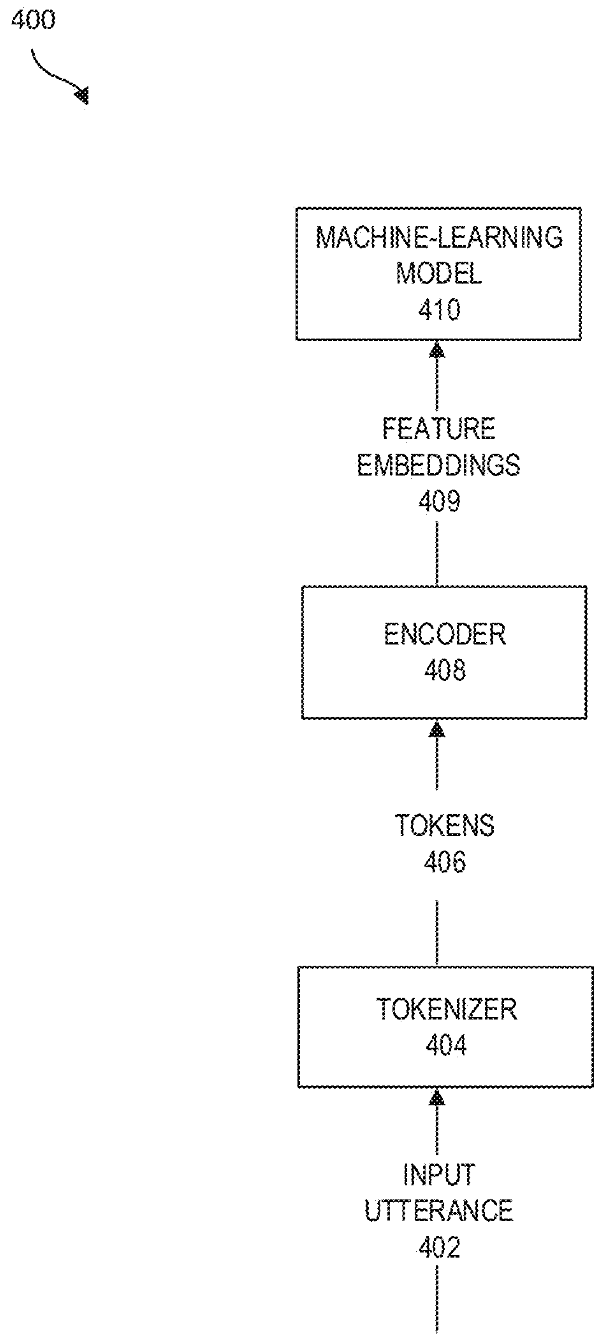

400
MACHINE-LEARNING MODEL 410
FEATURE EMBEDDINGS 409
ENCODER 408
TOKENS 406
TOKENIZER 404
INPUT UTTERANCE 402

FIG. 4A

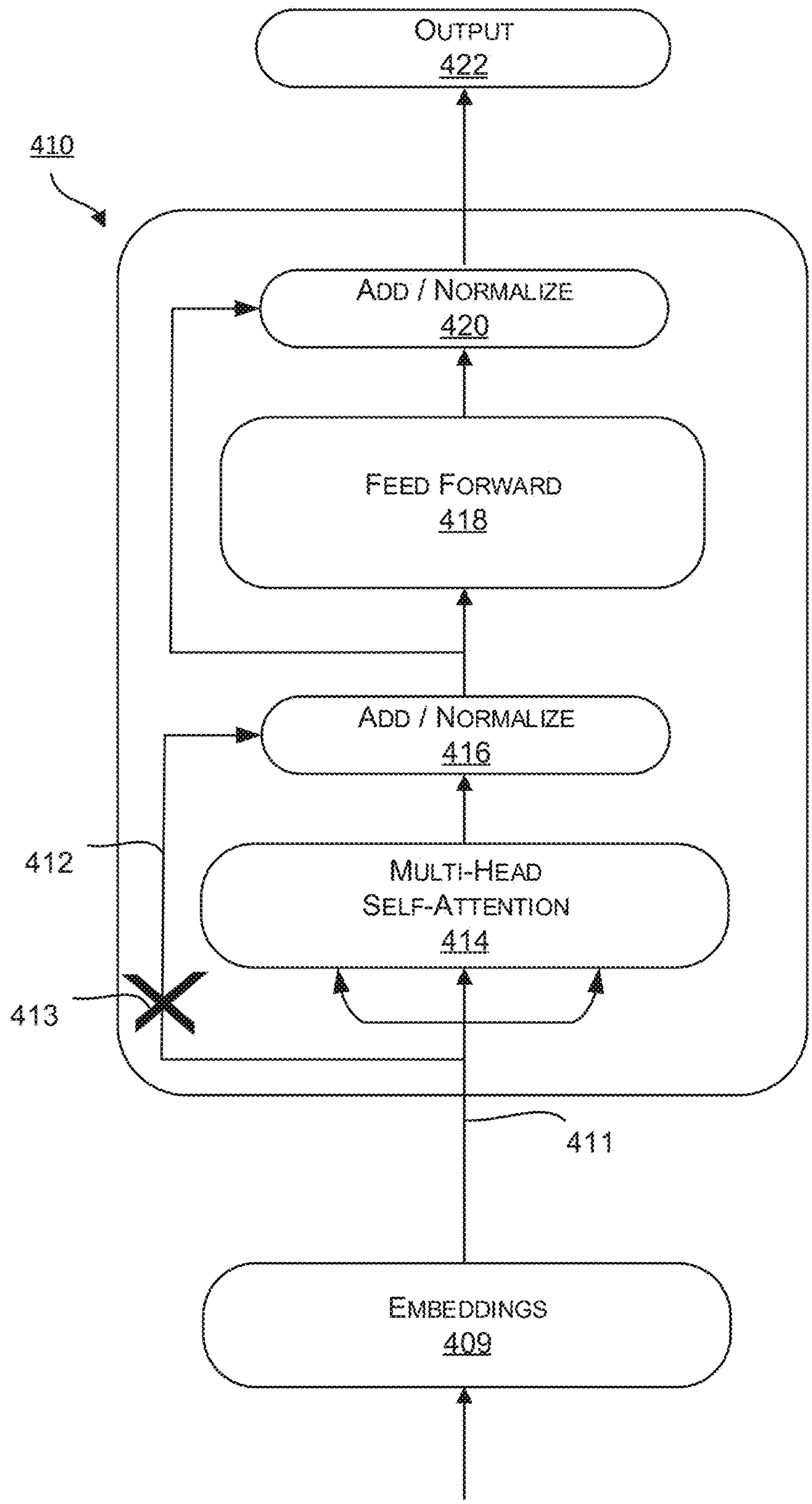

OUTPUT
422
410
ADD / NORMALIZE
420
FEED FORWARD
418
ADD / NORMALIZE
416
412
MULTI-HEAD
SELF-ATTENTION
414
413
411
EMBEDDINGS
409

FIG. 4B

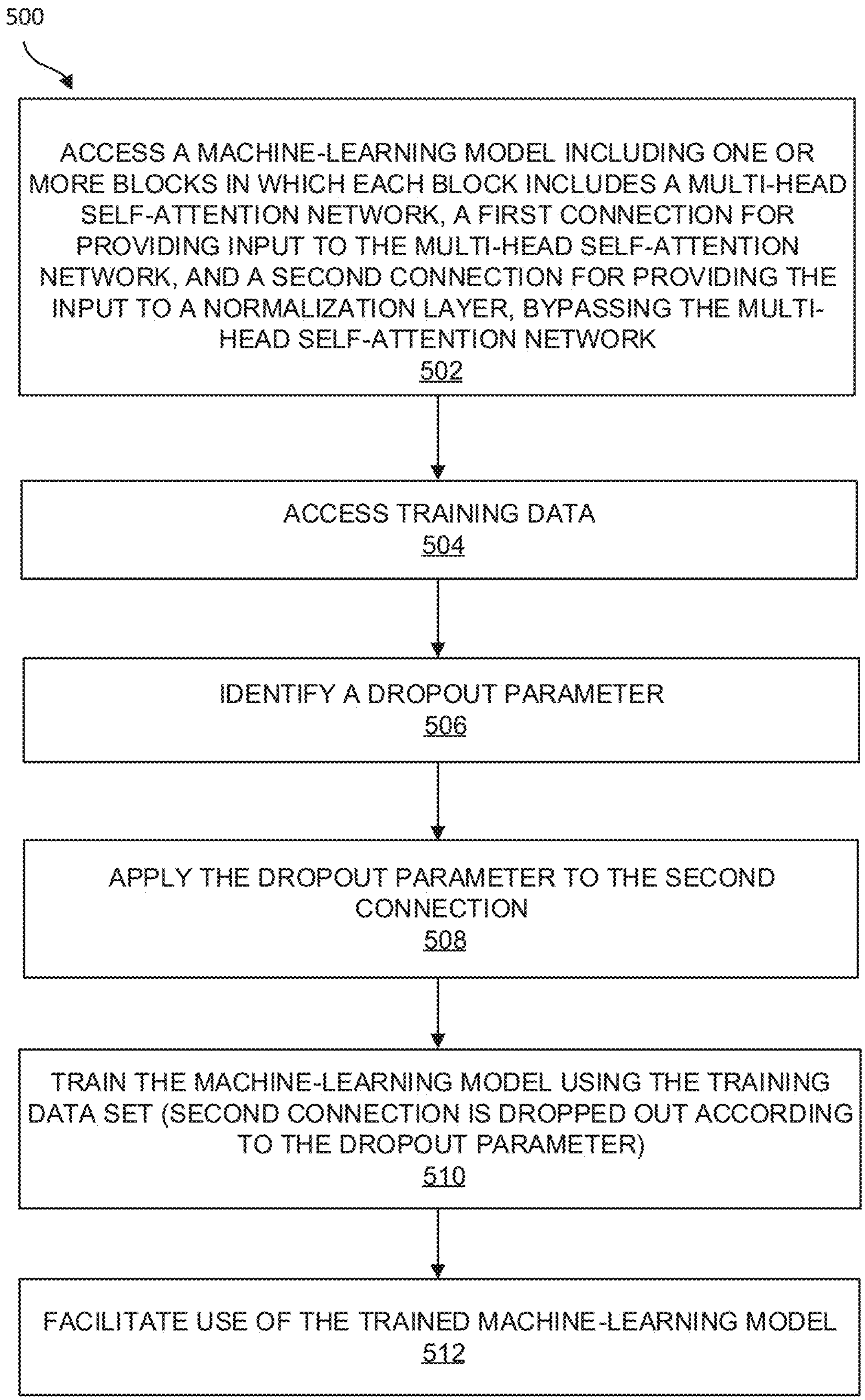

500
ACCESS A MACHINE-LEARNING MODEL INCLUDING ONE OR MORE BLOCKS IN WHICH EACH BLOCK INCLUDES A MULTI-HEAD SELF-ATTENTION NETWORK, A FIRST CONNECTION FOR PROVIDING INPUT TO THE MULTI-HEAD SELF-ATTENTION NETWORK, AND A SECOND CONNECTION FOR PROVIDING THE INPUT TO A NORMALIZATION LAYER, BYPASSING THE MULTI-HEAD SELF-ATTENTION NETWORK
502
ACCESS TRAINING DATA
504
IDENTIFY A DROPOUT PARAMETER
506
APPLY THE DROPOUT PARAMETER TO THE SECOND CONNECTION
508
TRAIN THE MACHINE-LEARNING MODEL USING THE TRAINING DATA SET (SECOND CONNECTION IS DROPPED OUT ACCORDING TO THE DROPOUT PARAMETER)
510
FACILITATE USE OF THE TRAINED MACHINE-LEARNING MODEL
512

FIG. 5A

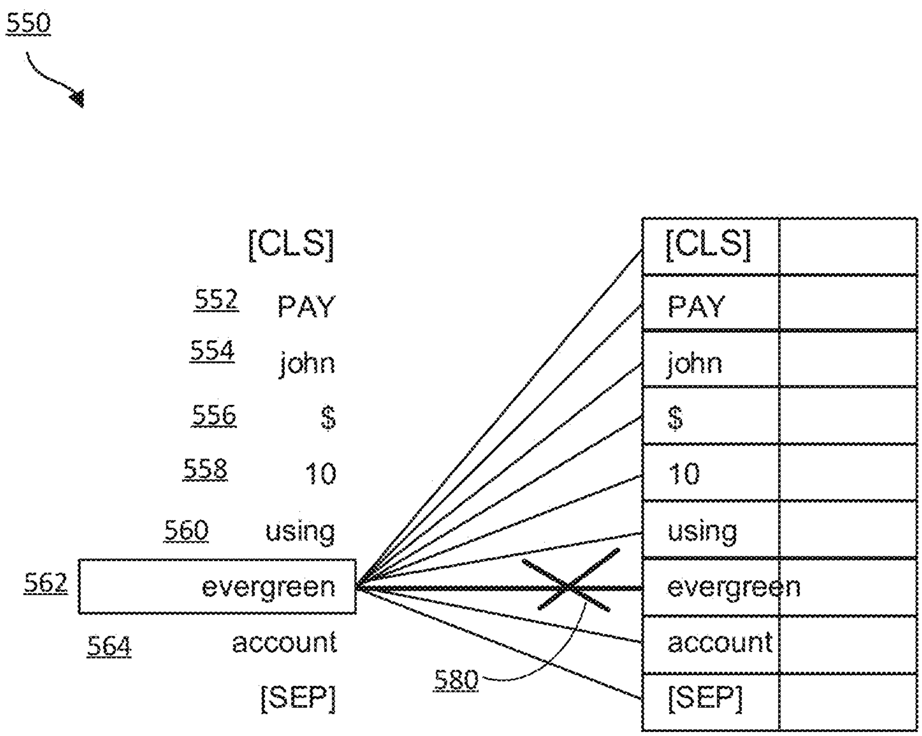

550
[CLS]
552 PAY
554 john
556 $
558 10
560 using
562 evergreen
564 account
[SEP]
[CLS]
PAY
john
$
10
using
evergreen
account
[SEP]
580

FIG. 5B

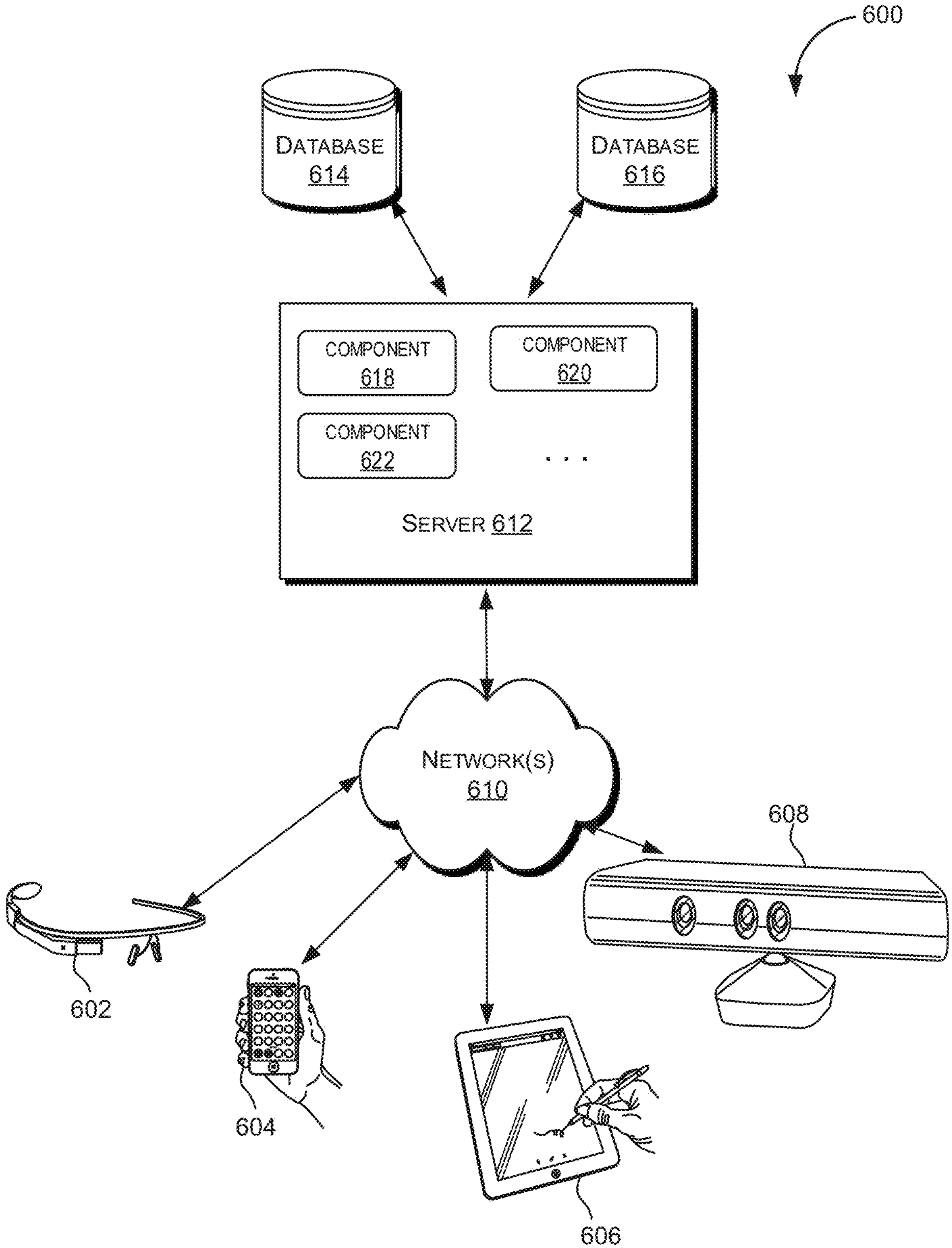

600
DATABASE
614
DATABASE
616
COMPONENT
618
COMPONENT
620
COMPONENT
622
. . .
SERVER 612
NETWORK(S)
610
608
602
604
606

FIG. 6

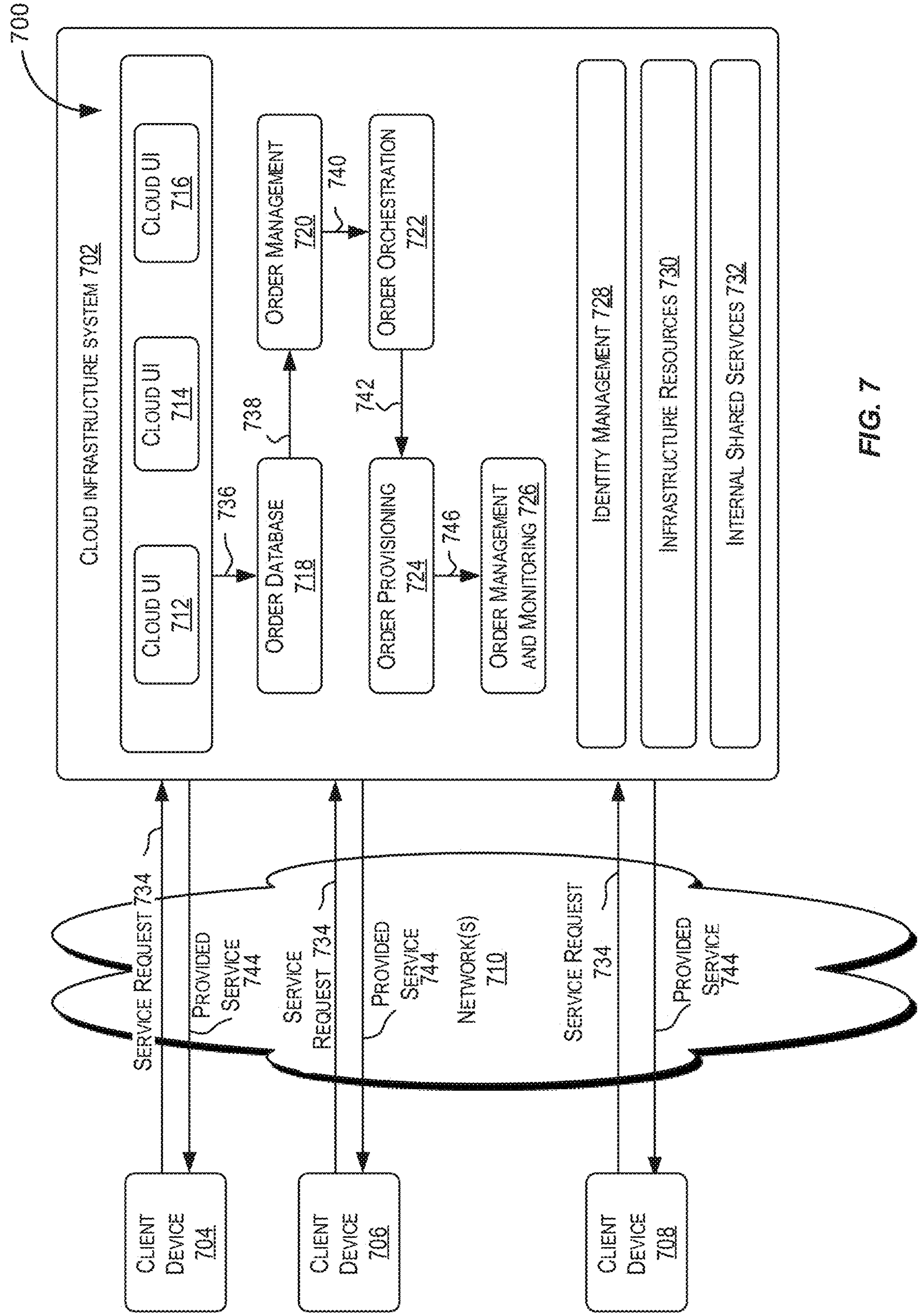

700
Cloud Infrastructure System 702
Cloud UI 712
Cloud UI 714
Cloud UI 716
736
Order Database 718
738
Order Management 720
740
Order Orchestration 722
742
Order Provisioning 724
746
Order Management and Monitoring 726
Identity Management 728
Infrastructure Resources 730
Internal Shared Services 732
Service Request 734
Provided Service 744
Service Request 734
Provided Service 744
Network(s) 710
Service Request 734
Provided Service 744
Client Device 704
Client Device 706
Client Device 708

FIG. 7

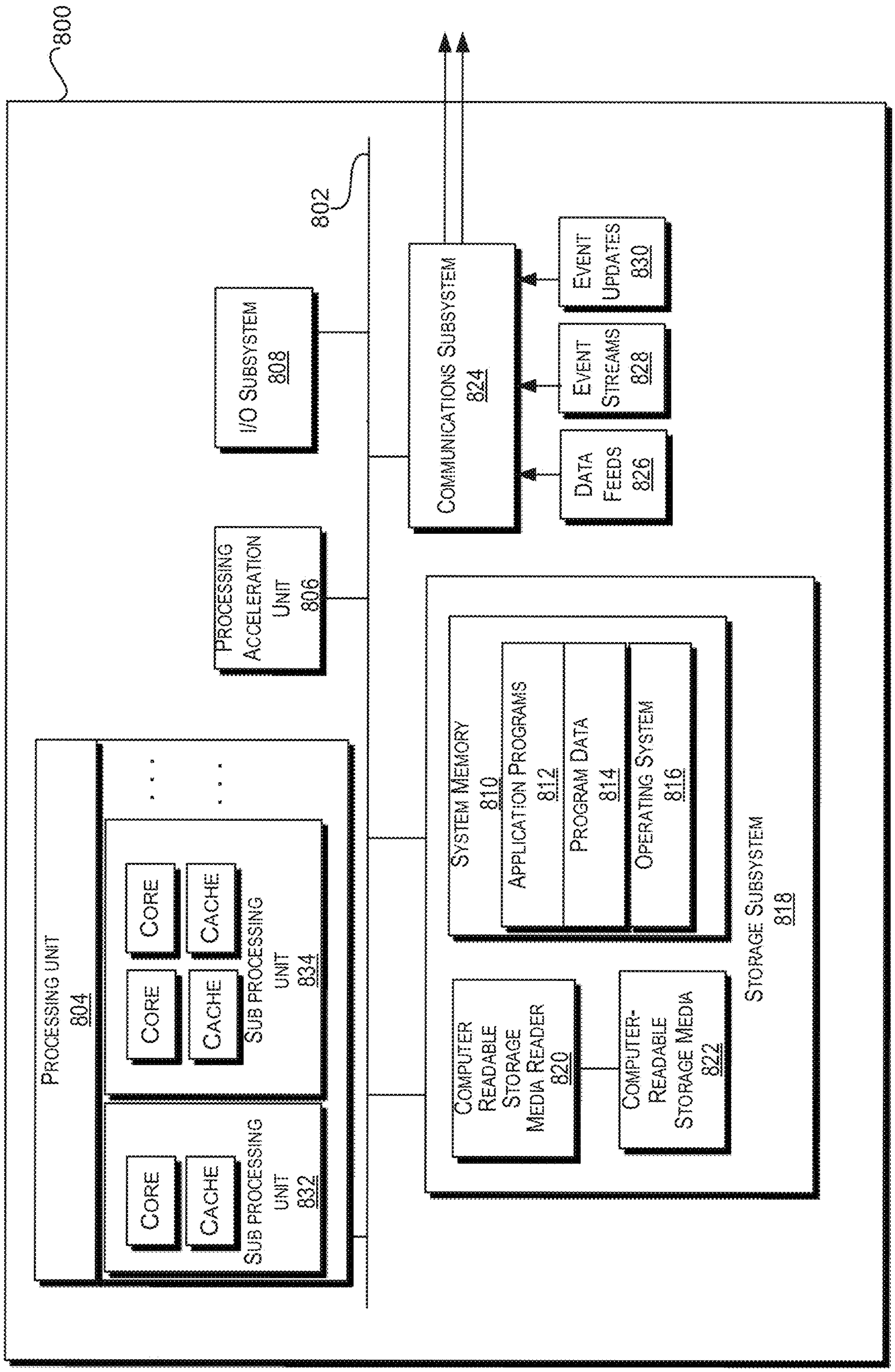

800
802
Processing Unit
804
Core
Cache
Sub Processing
Unit
832
Core
Core
Cache
Cache
Sub Processing
Unit
834
Processing
Acceleration
Unit
806
I/O Subsystem
808
Communications Subsystem
824
Data
Feeds
826
Event
Streams
828
Event
Updates
830
System Memory
810
Application Programs
812
Program Data
814
Operating System
816
Computer
Readable
Storage
Media Reader
820
Computer-
Readable
Storage Media
822
Storage Subsystem
818

FIG. 8

PATH DROPOUT FOR NATURAL LANGUAGE PROCESSING

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the priority and benefit of U.S. Provisional Application No. 63/280,580, filed on Nov. 17, 2021, the disclosure of which is incorporated herein by reference in its entirety for all purposes.

FIELD OF THE INVENTION

The present disclosure relates generally to artificial intelligence techniques, and more particularly, to techniques for training a machine learning model for natural language processing using path dropout.

BACKGROUND

Artificial intelligence has many applications. To illustrate, many users around the world are on instant messaging or chat platforms in order to get instant reactions. Organizations often use these instant messaging or chat platforms to engage with customers (or end users) in live conversations. However, it can be very costly for organizations to employ service people to engage in live communication with customers or end users. Chatbots or bots have begun to be developed to simulate conversations with end users, especially over the Internet. End users can communicate with bots through messaging apps that the end users have already installed and used. An intelligent bot, generally powered by artificial intelligence (AI), can communicate more intelligently and contextually in live conversations, and thus may allow for a more natural conversation between the bot and the end users for improved conversational experience. Instead of the end user learning a fixed set of keywords or commands that the bot knows how to respond to, an intelligent bot may be able to understand the end user's intention based upon user utterances in natural language and respond accordingly.

However, artificial intelligence-based solutions, such as chatbots, can be difficult to build because these automated solutions require specific knowledge in certain fields and the application of certain techniques that may be solely within the capabilities of specialized developers. As part of building such chatbots, a developer may first understand the needs of enterprises and end users. The developer may then analyze and make decisions related to, for example, selecting data sets to be used for the analysis, preparing the input data sets for analysis (e.g., cleansing the data, extracting, formatting, and/or transforming the data prior to analysis, performing data features engineering, etc.), identifying an appropriate machine learning (ML) technique(s) or model(s) for performing the analysis, and improving the technique or model to improve results/outcomes based upon feedback. The task of identifying an appropriate model may include developing multiple models, possibly in parallel, iteratively testing and experimenting with these models, before identifying a particular model (or models) for use. Further, supervised learning-based solutions typically involve a training phase, followed by an application (i.e., inference) phase, and iterative loops between the training phase and the application phase. The developer may be responsible for carefully implementing and monitoring these phases to achieve optimal solutions. For example, to train the ML technique(s) or model(s), precise training data is required to enable the algorithms to understand and learn certain patterns or features (e.g., for chatbots—intent extraction and careful syntactic analysis, not just raw language processing) that the ML technique(s) or model(s) will use to predict the outcome desired (e.g., inference of an intent from an utterance). In order to ensure the ML technique(s) or model(s) learn these pattern and features properly, the developer may be responsible for selecting, enriching, and optimizing sets of training data for the ML technique(s) or model(s).

BRIEF SUMMARY

Techniques disclosed herein relate generally to artificial intelligence techniques. More specifically and without limitation, techniques disclosed herein relate to techniques for improved training of a machine-learning model using dropout to teach the machine-learning model to focus on contextual features.

In various embodiments, a computer-implemented method is provided for training a machine-learning model to process audio or textual language input that includes accessing a machine-learning model, the machine-learning model including one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network; accessing a training data set; identifying a dropout parameter; applying the dropout parameter to the second connection; training the machine-learning model using the training data set to generate a trained machine-learning model, wherein the second connection is dropped out according to the dropout parameter; and facilitating use of the trained machine-learning model.

In some embodiments, the machine learning model includes a plurality of attention heads, each attention head providing an output that is concatenated and multiplied by an attention weight matrix; and the method further includes dropping out diagonal entries in the attention weight matrix.

In some embodiments, the dropout parameter is a hyperparameter of the machine-learning model, and the method further comprises performing hypertuning to identify the dropout parameter.

In some embodiments, when the second connection is dropped out, the machine-learning model is caused to learn contextual information through multi-headed self-attention. In some aspects, the training data set comprises one or more utterances, each utterance comprising one or more words that are labeled as an entity and one or more words that are not labeled as entities; and the contextual information is determined based on the one or more words that are not labeled as entities.

In some embodiments, facilitating use of the trained machine-learning model comprises inputting an utterance to the trained machine-learning model to identify an entity; and based on the identified entity, preparing and providing a response to the utterance.

In some embodiments, the dropout parameter is a dropout rate and the second connection is a residual connection, and applying the dropout parameter comprises dropping out the residual connection according to the dropout rate.

In various embodiments, a system is provided that includes one or more data processors and a non-transitory computer readable storage medium containing instructions which, when executed on the one or more data processors, cause the one or more data processors to perform operations comprising operations comprising accessing a machine-learning model, the machine-learning model including one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network; accessing a training data set; identifying a dropout parameter; applying the dropout parameter to the second connection; training the machine-learning model using the training data set to generate a trained machine-learning model, wherein the second connection is dropped out according to the dropout parameter; and facilitating use of the trained machine-learning model.

In various embodiments, a computer-program product is provided that is tangibly embodied in a non-transitory machine-readable storage medium and that includes instructions executable by one or more processors to cause the one or more processors to perform operations comprising accessing a machine-learning model, the machine-learning model including one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network; accessing a training data set; identifying a dropout parameter; applying the dropout parameter to the second connection; training the machine-learning model using the training data set to generate a trained machine learning model, wherein the second connection is dropped out according to the dropout parameter; and facilitating use of the trained machine-learning model.

The techniques described above and below may be implemented in a number of ways and in a number of contexts. Several example implementations and contexts are provided with reference to the following figures, as described below in more detail. However, the following implementations and contexts are but a few of many.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a simplified block diagram of a distributed environment incorporating an exemplary embodiment.

FIG. **2** is a simplified block diagram of a computing system implementing a master bot according to certain embodiments.

FIG. **3** is a simplified block diagram of a computing system implementing a skill bot according to certain embodiments.

FIG. **4**A is a simplified block diagram of a chatbot training and deployment system in accordance with various embodiments.

FIG. **4**B is a simplified block diagram of a machine-learning model in accordance with various embodiments.

FIG. **5**A is a flowchart illustrating a process for generating and using a machine learning model with path dropout according to certain embodiments.

FIG. **5**B is a diagram illustrating an example of token pairings with dropout, according to some embodiments.

FIG. **6** depicts a simplified diagram of a distributed system for implementing various embodiments.

FIG. **7** is a simplified block diagram of one or more components of a system environment by which services provided by one or more components of an embodiment system may be offered as cloud services, in accordance with various embodiments.

FIG. **8** illustrates an example computer system that may be used to implement various embodiments.

DETAILED DESCRIPTION

In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of certain embodiments. However, it will be apparent that various embodiments may be practiced without these specific details. The figures and description are not intended to be restrictive. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any embodiment or design described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or designs.

INTRODUCTION

Artificial intelligence has many applications. For example, a digital assistant is an artificial intelligent driven interface that helps users accomplish a variety of tasks in natural language conversations. For each digital assistant, a customer may assemble one or more skills. Skills (also described herein as chatbots, bots, or skill bots) are individual bots that are focused on specific types of tasks, such as tracking inventory, submitting time cards, and creating expense reports. When an end user engages with the digital assistant, the digital assistant evaluates the end user input and routes the conversation to and from the appropriate chatbot. The digital assistant can be made available to end users through a variety of channels such as FACEBOOK® Messenger, SKYPE MOBILE® messenger, or a Short Message Service (SMS). Channels carry the chat back and forth from end users on various messaging platforms to the digital assistant and its various chatbots. The channels may also support user agent escalation, event-initiated conversations, and testing.

A chatbot processes an input utterance to discern meaning. As used herein, an utterance or a message may refer to a set of words (e.g., one or more sentences) exchanged during a conversation with a chatbot.

As part of the processing performed by artificial intelligence-based technology such as a chatbot system, named entity recognition may be performed by a Named Entity Recognizer (NER) component. NERs are models trained to recognize the names of things, which can be proper names (e.g., Florida), goods (e.g., shoes), services (e.g., massage), and so forth. The NERs may also be trained to recognize descriptive words or phrases such as colors and sizes. Accordingly, "named entity" or "entity" as used herein can include words or phrases that a machine learning model has been trained to identify, including things, as well as words or phrases associated with such things. Upon recognizing a named entity, the NER may label the corresponding word or phrase with a named entity type. A named entity type is a category of named entity. For example, the phrase New York is labeled with the named entity type city, and the word seventy is labeled with the named entity type temperature. The NER includes a machine learning model (e.g., neural networks) trained to identify a named entity corresponding to identified words or phrases.

Building an artificial intelligence-based system such as a chatbot system that can accurately identify a named entity based upon user utterances is a challenging task in part due to the subtleties and ambiguity of natural languages and the dimension of the input space (e.g., possible user utterances, small size of training data) and the size of the output space (number of named entities). As an example, in the training process, the NER model may place excessive focus on the entity value itself rather than the surrounding contextual information. As a specific example, in the utterance "I want to go to Central Park," "Central Park" is a named entity and the rest of the utterance, "I want to go to," provides contextual information useable for named entity recognition and other dialog processing tasks. It has been observed that, when performing named entity recognition using traditional techniques, some entities are misclassified due to insufficient attention on contextual information.

When performing named entity recognition, it has been found that often the model does not pay much attention to the context. For example, for the utterance "Pay John $10 using Evergreen account," the model should tag "Evergreen" as the entity type MerchantType since Evergreen is followed by the word "account," which is a strong context signal. Illustrative results are shown in Table 1.

TABLE 1

EXAMPLE NER RESULTS WITHOUT PATH DROPOUT

query	expected Entities	detect Entities
what is the total amount I spent at W H Smiths	MERCHANT_NAME: W H Smiths	MERCHANT_NAME: Smiths
$40 for office supplies from Office Depot	MERCHANT_NAME: Office Depot	MERCHANT_NAME: Office
Had drink with client at Boba Guys for $9.15	MERCHANT_NAME: Boba Guys	MERCHANT_NAME: Boba

In Table 1, the first column is a query or input utterance, and the second column shows the target named entities to be identified. The third column shows examples of incorrect named entity tags generated using prior techniques (e.g., without the dropout techniques described herein). For example, when the target named entity is the merchant name "Home Depot," the merchant name is incorrectly identified as Depot due to lack of attention to the relevant contextual data. One reason that may explain why the model behaves like this is that it focuses too much on lexical features during training time.

In some embodiments, to cause the model to focus more on contextual information, path dropout is implemented. Path dropout can be achieved by dropping out the paths from a word/subword embedding to its representation output. More specifically, the system applies dropout to (a) the second connection **412** shown in FIG. **4**B and/or (b) the diagonal entries in an attention weight matrix. By doing this, the residual path from a word/subword input embedding to its output representation is blocked. Thus, the model will need to rely on context information (e.g., through a multi-head self-attention network, as described herein).

The dropout techniques of the present disclosure are used to train a machine-learning model to focus on contextual information more, which provides improved results. The machine-learning model can be trained to perform named entity recognition or other natural language processing tasks. Dropout involves randomly dropping out, or ignoring, certain layer outputs and/or connections. In some embodiments, a particular residual connection in the machine-learning model is dropped out to enforce context learning.

In some embodiments, the machine-learning model is or includes a transformer block, which may be part of a pretrained language model. The transformer block includes a multi-head self-attention network. The multi-head self-attention network receives input embeddings via a first connection (e.g., to a previous encoder). The machine-learning model includes a residual connection for providing the input embeddings to a normalization layer, bypassing the multi-head self-attention network. Path dropout is applied to drop out the residual connection, which increases the likelihood that self-attention will be applied to discern contextual information.

In some embodiments, dropout is further applied to an attention weight matrix, which further enforces contextual learning and improves named entity recognition performance. For example, the machine learning model includes multiple attention heads, each attention head providing an output that is concatenated and multiplied by an attention weight matrix. Selected entries in the attention weight matrix are dropped out randomly. In some aspects, diagonal entries in the attention weight matrix are dropped out, which corresponds to ignoring entries pairing a token to itself. Instead, entries pairing a token with other contextual tokens are kept, which also causes the model to increasingly focus on contextual information.

In some embodiments, the dropout is applied using a dropout parameter, which is a hyperparameter of the machine-learning model. A hypertuning process may be executed to select an appropriate dropout rate, which controls how frequently the path is dropped out.

Bot and Analytic Systems

A bot (also referred to as a skill, chatbot, chatterbot, or talkbot) is a computer program that can perform conversations with end users. The bot can generally respond to natural-language messages (e.g., questions or comments) through a messaging application that uses natural-language messages. Enterprises may use one or more bot systems to communicate with end users through a messaging application. The messaging application, which may be referred to as a channel, may be an end user preferred messaging application that the end user has already installed and familiar with. Thus, the end user does not need to download and install new applications in order to chat with the bot system. The messaging application may include, for example, over-the-top (OTT) messaging channels (such as Facebook Messenger, Facebook WhatsApp, WeChat, Line, Kik, Telegram, Talk, Skype, Slack, or SMS), virtual private assistants (such as Amazon Dot, Echo, or Show, Google Home, Apple HomePod, etc.), mobile and web app extensions that extend native or hybrid/responsive mobile apps or web applications with chat capabilities, or voice based input (such as devices or apps with interfaces that use Siri, Cortana, Google Voice, or other speech input for interaction).

In some examples, a bot system may be associated with a Uniform Resource Identifier (URI). The URI may identify the bot system using a string of characters. The URI may be used as a webhook for one or more messaging application systems. The URI may include, for example, a Uniform Resource Locator (URL) or a Uniform Resource Name (URN). The bot system may be designed to receive a message (e.g., a hypertext transfer protocol (HTTP) post call message) from a messaging application system. The HTTP post call message may be directed to the URI from the messaging application system. In some embodiments, the message may be different from a HTTP post call message. For example, the bot system may receive a message from a Short Message Service (SMS). While discussion herein may refer to communications that the bot system receives as a message, it should be understood that the message may be an HTTP post call message, a SMS message, or any other type of communication between two systems.

End users may interact with the bot system through a conversational interaction (sometimes referred to as a conversational user interface (UI)), just as interactions between people. In some cases, the interaction may include the end user saying "Hello" to the bot and the bot responding with a "Hi" and asking the end user how it can help. In some cases, the interaction may also be a transactional interaction with, for example, a banking bot, such as transferring money from one account to another; an informational interaction with, for example, a HR bot, such as checking for vacation balance; or an interaction with, for example, a retail bot, such as discussing returning purchased goods or seeking technical support.

In some embodiments, the bot system may intelligently handle end user interactions without interaction with an administrator or developer of the bot system. For example, an end user may send one or more messages to the bot system in order to achieve a desired goal. A message may include certain content, such as text, emojis, audio, image, video, or other method of conveying a message. In some embodiments, the bot system may convert the content into a standardized form (e.g., a representational state transfer (REST) call against enterprise services with the proper parameters) and generate a natural language response. The bot system may also prompt the end user for additional input parameters or request other additional information. In some embodiments, the bot system may also initiate communication with the end user, rather than passively responding to end user utterances. Described herein are various techniques for identifying an explicit invocation of a bot system and determining an input for the bot system being invoked. In certain embodiments, explicit invocation analysis is performed by a master bot based on detecting an invocation name in an utterance. In response to detection of the invocation name, the utterance may be refined for input to a skill bot associated with the invocation name.

A conversation with a bot may follow a specific conversation flow including multiple states. The flow may define what would happen next based on an input. In some embodiments, a state machine that includes user defined states (e.g., end user intents) and actions to take in the states or from state to state may be used to implement the bot system. A conversation may take different paths based on the end user input, which may impact the decision the bot makes for the flow. For example, at each state, based on the end user input or utterances, the bot may determine the end user's intent in order to determine the appropriate next action to take. As used herein and in the context of an utterance, the term "intent" refers to an intent of the user who provided the utterance. For example, the user may intend to engage a bot in conversation for ordering pizza, so that the user's intent could be represented through the utterance "Order pizza." A user intent can be directed to a particular task that the user wishes a chatbot to perform on behalf of the user. Therefore, utterances can be phrased as questions, commands, requests, and the like, that reflect the user's intent. An intent may include a goal that the end user would like to accomplish.

In the context of the configuration of a chatbot, the term "intent" is used herein to refer to configuration information for mapping a user's utterance to a specific task/action or category of task/action that the chatbot can perform. In order to distinguish between the intent of an utterance (i.e., a user intent) and the intent of a chatbot, the latter is sometimes referred to herein as a "bot intent." A bot intent may comprise a set of one or more utterances associated with the intent. For instance, an intent for ordering pizza can have various permutations of utterances that express a desire to place an order for pizza. These associated utterances can be used to train an intent classifier of the chatbot to enable the intent classifier to subsequently determine whether an input utterance from a user matches the order pizza intent. A bot intent may be associated with one or more dialog flows for starting a conversation with the user and in a certain state. For example, the first message for the order pizza intent could be the question "What kind of pizza would you like?" In addition to associated utterances, a bot intent may further comprise named entities that relate to the intent. For example, the order pizza intent could include variables or parameters used to perform the task of ordering pizza, e.g., topping **1**, topping **2**, pizza type, pizza size, pizza quantity, and the like. The value of an entity is typically obtained through conversing with the user.

FIG. **1** is a simplified block diagram of an environment **100** incorporating a chatbot system according to certain embodiments. Environment **100** comprises a digital assistant builder platform (DABP) **102** that enables users of DABP **102** to create and deploy digital assistants or chatbot systems. DABP **102** can be used to create one or more digital assistants (or DAs) or chatbot systems. For example, as shown in FIG. **1**, user **104** representing a particular enterprise can use DABP **102** to create and deploy a digital assistant **106** for users of the particular enterprise. For example, DABP **102** can be used by a bank to create one or more digital assistants for use by the bank's customers. The same DABP **102** platform can be used by multiple enterprises to create digital assistants. As another example, an owner of a restaurant (e.g., a pizza shop) may use DABP **102** to create and deploy a digital assistant that enables customers of the restaurant to order food (e.g., order pizza).

For purposes of this disclosure, a "digital assistant" is an entity that helps users of the digital assistant accomplish various tasks through natural language conversations. A digital assistant can be implemented using software only (e.g., the digital assistant is a digital entity implemented using programs, code, or instructions executable by one or more processors), using hardware, or using a combination of hardware and software. A digital assistant can be embodied or implemented in various physical systems or devices, such as in a computer, a mobile phone, a watch, an appliance, a vehicle, and the like. A digital assistant is also sometimes referred to as a chatbot system. Accordingly, for purposes of this disclosure, the terms digital assistant and chatbot system are interchangeable.

A digital assistant, such as digital assistant **106** built using DABP **102**, can be used to perform various tasks via natural language-based conversations between the digital assistant and its users **108**. As part of a conversation, a user may provide one or more user inputs **110** to digital assistant **106** and get responses **112** back from digital assistant **106**. A conversation can include one or more of inputs **110** and responses **112**. Via these conversations, a user can request one or more tasks to be performed by the digital assistant and, in response, the digital assistant is configured to perform the user-requested tasks and respond with appropriate responses to the user.

User inputs **110** are generally in a natural language form and are referred to as utterances. A user utterance **110** can be in text form, such as when a user types in a sentence, a question, a text fragment, or even a single word and provides it as input to digital assistant **106**. In some embodiments, a user utterance **110** can be in audio input or speech form, such as when a user says or speaks something that is provided as input to digital assistant **106**. The utterances are typically in a language spoken by the user **108**. For example, the utterances may be in English, or some other language. When an utterance is in speech form, the speech input is converted to text form utterances in that particular language and the text utterances are then processed by digital assistant **106**. Various speech-to-text processing techniques may be used to convert a speech or audio input to a text utterance, which is then processed by digital assistant **106**. In some embodiments, the speech-to-text conversion may be done by digital assistant **106** itself.

An utterance, which may be a text utterance or a speech utterance, can be a fragment, a sentence, multiple sentences, one or more words, one or more questions, combinations of the aforementioned types, and the like. Digital assistant **106** is configured to apply natural language understanding (NLU) techniques to the utterance to understand the meaning of the user input. As part of the NLU processing for a utterance, digital assistant **106** is configured to perform processing to understand the meaning of the utterance, which involves identifying one or more intents and one or more entities corresponding to the utterance. Upon understanding the meaning of an utterance, digital assistant **106** may perform one or more actions or operations responsive to the understood meaning or intents. For purposes of this disclosure, it is assumed that the utterances are text utterances that have been provided directly by a user **108** of digital assistant **106** or are the results of conversion of input speech utterances to text form. This however is not intended to be limiting or restrictive in any manner.

For example, a user **108** input may request a pizza to be ordered by providing an utterance such as "I want to order a pizza." Upon receiving such an utterance, digital assistant **106** is configured to understand the meaning of the utterance and take appropriate actions. The appropriate actions may involve, for example, responding to the user with questions requesting user input on the type of pizza the user desires to order, the size of the pizza, any toppings for the pizza, and the like. The responses provided by digital assistant **106** may also be in natural language form and typically in the same language as the input utterance. As part of generating these responses, digital assistant **106** may perform natural language generation (NLG). For the user ordering a pizza, via the conversation between the user and digital assistant **106**, the digital assistant may guide the user to provide all the requisite information for the pizza order, and then at the end of the conversation cause the pizza to be ordered. Digital assistant **106** may end the conversation by outputting information to the user indicating that the pizza has been ordered.

At a conceptual level, digital assistant **106** performs various processing in response to an utterance received from a user. In some embodiments, this processing involves a series or pipeline of processing steps including, for example, understanding the meaning of the input utterance (sometimes referred to as Natural Language Understanding (NLU), determining an action to be performed in response to the utterance, where appropriate causing the action to be performed, generating a response to be output to the user responsive to the user utterance, outputting the response to the user, and the like. The NLU processing can include parsing the received input utterance to understand the structure and meaning of the utterance, refining and reforming the utterance to develop a better understandable form (e.g., logical form) or structure for the utterance. Generating a response may include using NLG techniques.

The NLU processing performed by a digital assistant, such as digital assistant **106**, can include various NLP related processing such as sentence parsing (e.g., tokenizing, lemmatizing, identifying part-of-speech tags for the sentence, identifying named entities in the sentence, generating dependency trees to represent the sentence structure, splitting a sentence into clauses, analyzing individual clauses, resolving anaphoras, performing chunking, and the like). In certain embodiments, the NLU processing or portions thereof is performed by digital assistant **106** itself. In some other embodiments, digital assistant **106** may use other resources to perform portions of the NLU processing. For example, the syntax and structure of an input utterance sentence may be identified by processing the sentence using a parser, a part-of-speech tagger, and/or a named entity recognizer. In one implementation, for the English language, a parser, a part-of-speech tagger, and a named entity recognizer such as ones provided by the Stanford Natural Language Processing (NLP) Group are used for analyzing the sentence structure and syntax. These are provided as part of the Stanford CoreNLP toolkit.

While the various examples provided in this disclosure show utterances in the English language, this is meant only as an example. In certain embodiments, digital assistant **106** is also capable of handling utterances in languages other than English. Digital assistant **106** may provide subsystems (e.g., components implementing NLU functionality) that are configured for performing processing for different languages. These subsystems may be implemented as pluggable units that can be called using service calls from an NLU core server. This makes the NLU processing flexible and extensible for each language, including allowing different orders of processing. A language pack may be provided for individual languages, where a language pack can register a list of subsystems that can be served from the NLU core server.

A digital assistant, such as digital assistant **106** depicted in FIG. **1**, can be made available or accessible to its users **108** through a variety of different channels, such as but not limited to, via certain applications, via social media platforms, via various messaging services and applications, and other applications or channels. A single digital assistant can have several channels configured for it so that it can be run on and be accessed by different services simultaneously.

A digital assistant or chatbot system generally contains or is associated with one or more skills. In certain embodiments, these skills are individual chatbots (referred to as skill bots) that are configured to interact with users and fulfill specific types of tasks, such as tracking inventory, submitting timecards, creating expense reports, ordering food, checking a bank account, making reservations, buying a widget, and the like. For example, for the embodiment depicted in FIG. **1**, digital assistant or chatbot system **106** includes skills **116-1**, **116-2**, and so on. For purposes of this disclosure, the terms "skill" and "skills" are used synonymously with the terms "skill bot" and "skill bots," respectively.

Each skill associated with a digital assistant helps a user of the digital assistant complete a task through a conversation with the user, where the conversation can include a combination of text or audio inputs provided by the user and responses provided by the skill bots. These responses may be in the form of text or audio messages to the user and/or using simple user interface elements (e.g., select lists) that are presented to the user for the user to make selections.

There are various ways in which a skill or skill bot can be associated or added to a digital assistant. In some instances, a skill bot can be developed by an enterprise and then added to a digital assistant using DABP **102**. In other instances, a skill bot can be developed and created using DABP **102** and then added to a digital assistant created using DABP **102**. In yet other instances, DABP **102** provides an online digital store (referred to as a "skills store") that offers multiple skills directed to a wide range of tasks. The skills offered through the skills store may also expose various cloud services. In order to add a skill to a digital assistant being generated using DABP **102**, a user of DABP **102** can access the skills store via DABP **102**, select a desired skill, and indicate that the selected skill is to be added to the digital assistant created using DABP **102**. A skill from the skills store can be added to a digital assistant as is or in a modified form (for example, a user of DABP **102** may select and clone a particular skill bot provided by the skills store, make customizations or modifications to the selected skill bot, and then add the modified skill bot to a digital assistant created using DABP **102**).

Various different architectures may be used to implement a digital assistant or chatbot system. For example, in certain embodiments, the digital assistants created and deployed using DABP **102** may be implemented using a master bot/child(or sub) bot paradigm or architecture. According to this paradigm, a digital assistant is implemented as a master bot that interacts with one or more child bots that are skill bots. For example, in the embodiment depicted in FIG. **1**, digital assistant **106** comprises a master bot **114** and skill bots **116-1**, **116-2**, etc. that are child bots of master bot **114**. In certain embodiments, digital assistant **106** is itself considered to act as the master bot.

A digital assistant implemented according to the master-child bot architecture enables users of the digital assistant to interact with multiple skills through a unified user interface, namely via the master bot. When a user engages with a digital assistant, the user input is received by the master bot. The master bot then performs processing to determine the meaning of the user input utterance. The master bot then determines whether the task requested by the user in the utterance can be handled by the master bot itself, else the master bot selects an appropriate skill bot for handling the user request and routes the conversation to the selected skill bot. This enables a user to converse with the digital assistant through a common single interface and still provide the capability to use several skill bots configured to perform specific tasks. For example, for a digital assistance developed for an enterprise, the master bot of the digital assistant may interface with skill bots with specific functionalities, such as a CRM bot for performing functions related to customer relationship management (CRM), an ERP bot for performing functions related to enterprise resource planning (ERP), an HCM bot for performing functions related to human capital management (HCM), etc. This way the end user or consumer of the digital assistant need only know how to access the digital assistant through the common master bot interface and behind the scenes multiple skill bots are provided for handling the user request.

In certain embodiments, in a master bot/child bots infrastructure, the master bot is configured to be aware of the available list of skill bots. The master bot may have access to metadata that identifies the various available skill bots, and for each skill bot, the capabilities of the skill bot including the tasks that can be performed by the skill bot. Upon receiving a user request in the form of an utterance, the master bot is configured to, from the multiple available skill bots, identify or predict a specific skill bot that can best serve or handle the user request. The master bot then routes the utterance (or a portion of the utterance) to that specific skill bot for further handling. Control thus flows from the master bot to the skill bots. The master bot can support multiple input and output channels. In certain embodiments, routing may be performed with the aid of processing performed by one or more available skill bots. For example, as discussed below, a skill bot can be trained to infer an intent for an utterance and to determine whether the inferred intent matches an intent with which the skill bot is configured. Thus, the routing performed by the master bot can involve the skill bot communicating to the master bot an indication of whether the skill bot has been configured with an intent suitable for handling the utterance.

While the embodiment in FIG. **1** shows digital assistant **106** comprising a master bot **114** and skill bots **116-1**, **116-2**, and **116-3**, this is not intended to be limiting. A digital assistant can include various other components (e.g., other systems and subsystems) that provide the functionalities of the digital assistant. These systems and subsystems may be implemented only in software (e.g., code, instructions stored on a computer-readable medium and executable by one or more processors), in hardware only, or in implementations that use a combination of software and hardware.

DABP **102** provides an infrastructure and various services and features that enable a user of DABP **102** to create a digital assistant including one or more skill bots associated with the digital assistant. In some instances, a skill bot can be created by cloning an existing skill bot, for example, cloning a skill bot provided by the skills store. As previously indicated, DABP **102** provides a skills store or skills catalog that offers multiple skill bots for performing various tasks. A user of DABP **102** can clone a skill bot from the skills store. As needed, modifications or customizations may be made to the cloned skill bot. In some other instances, a user of DABP **102** created a skill bot from scratch using tools and services offered by DABP **102**. As previously indicated, the skills store or skills catalog provided by DABP **102** may offer multiple skill bots for performing various tasks.

In certain embodiments, at a high level, creating or customizing a skill bot involves the following steps:

(1) Configuring settings for a new skill bot

(2) Configuring one or more intents for the skill bot

(3) Configuring one or more entities for one or more intents

(4) Training the skill bot

(5) Creating a dialog flow for the skill bot

(6) Adding custom components to the skill bot as needed

(7) Testing and deploying the skill bot

Each of the above steps is briefly described below.

(1) Configuring settings for a new skill bot—Various settings may be configured for the skill bot. For example, a skill bot designer can specify one or more invocation names for the skill bot being created. These invocation names can then be used by users of a digital assistant to explicitly invoke the skill bot. For example, a user can input an invocation name in the user's utterance to explicitly invoke the corresponding skill bot.

(2) Configuring one or more intents and associated example utterances for the skill bot—The skill bot designer specifies one or more intents (also referred to as bot intents) for a skill bot being created. The skill bot is then trained based upon these specified intents. These intents represent categories or classes that the skill bot is trained to infer for input utterances. Upon receiving an utterance, a trained skill bot infers an intent for the utterance, where the inferred intent is selected from the predefined set of intents used to train the skill bot. The skill bot then takes an appropriate action responsive to an utterance based upon the intent inferred for that utterance. In some instances, the intents for

a skill bot represent tasks that the skill bot can perform for users of the digital assistant. Each intent is given an intent identifier or intent name. For example, for a skill bot trained for a bank, the intents specified for the skill bot may include "CheckBalance," "TransferMoney," "DepositCheck," and the like.

For each intent defined for a skill bot, the skill bot designer may also provide one or more example utterances that are representative of and illustrate the intent. These example utterances are meant to represent utterances that a user may input to the skill bot for that intent. For example, for the CheckBalance intent, example utterances may include "What's my savings account balance?", "How much is in my checking account?", "How much money do I have in my account," and the like. Accordingly, various permutations of typical user utterances may be specified as example utterances for an intent.

The intents and the their associated example utterances are used as training data to train the skill bot. Various different training techniques may be used. As a result of this training, a predictive model is generated that is configured to take an utterance as input and output an intent inferred for the utterance by the predictive model. In some instances, input utterances are provided to an intent analysis engine, which is configured to use the trained model to predict or infer an intent for the input utterance. The skill bot may then take one or more actions based upon the inferred intent.

(3) Configuring entities for one or more intents of the skill bot—In some instances, additional context may be needed to enable the skill bot to properly respond to a user utterance. For example, there may be situations where a user input utterance resolves to the same intent in a skill bot. For instance, in the above example, utterances "What's my savings account balance?" and "How much is in my checking account?" both resolve to the same CheckBalance intent, but these utterances are different requests asking for different things. To clarify such requests, one or more entities are added to an intent. Using the banking skill bot example, an entity called AccountType, which defines values called "checking" and "saving" may enable the skill bot to parse the user request and respond appropriately. In the above example, while the utterances resolve to the same intent, the value associated with the AccountType entity is different for the two utterances. This enables the skill bot to perform possibly different actions for the two utterances in spite of them resolving to the same intent. One or more entities can be specified for certain intents configured for the skill bot. Entities are thus used to add context to the intent itself. Entities help describe an intent more fully and enable the skill bot to complete a user request.

In certain embodiments, there are two types of entities: (a) built-in entities provided by DABP **102**, and (2) custom entities that can be specified by a skill bot designer. Built-in entities are generic entities that can be used with a wide variety of bots. Examples of built-in entities include, without limitation, entities related to time, date, addresses, numbers, email addresses, duration, recurring time periods, currencies, phone numbers, URLs, and the like. Custom entities are used for more customized applications. For example, for a banking skill, an AccountType entity may be defined by the skill bot designer that enables various banking transactions by checking the user input for keywords like checking, savings, and credit cards, etc.

(4) Training the skill bot—A skill bot is configured to receive user input in the form of utterances parse or otherwise process the received input, and identify or select an intent that is relevant to the received user input. As indicated above, the skill bot has to be trained for this. In certain embodiments, a skill bot is trained based upon the intents configured for the skill bot and the example utterances associated with the intents (collectively, the training data), so that the skill bot can resolve user input utterances to one of its configured intents. In certain embodiments, the skill bot uses a predictive model that is trained using the training data and allows the skill bot to discern what users say (or in some cases, are trying to say). DABP **102** provides various different training techniques that can be used by a skill bot designer to train a skill bot, including various machine-learning based training techniques, rules-based training techniques, and/or combinations thereof. In certain embodiments, a portion (e.g., 80%) of the training data is used to train a skill bot model and another portion (e.g., the remaining 20%) is used to test or verify the model. Once trained, the trained model (also sometimes referred to as the trained skill bot) can then be used to handle and respond to user utterances. In certain cases, a user's utterance may be a question that requires only a single answer and no further conversation. In order to handle such situations, a Q&A (question-and-answer) intent may be defined for a skill bot. This enables a skill bot to output replies to user requests without having to update the dialog definition. Q&A intents are created in a similar manner as regular intents. The dialog flow for Q&A intents can be different from that for regular intents.

(5) Creating a dialog flow for the skill bot—A dialog flow specified for a skill bot describes how the skill bot reacts as different intents for the skill bot are resolved responsive to received user input. The dialog flow defines operations or actions that a skill bot will take, e.g., how the skill bot responds to user utterances, how the skill bot prompts users for input, how the skill bot returns data. A dialog flow is like a flowchart that is followed by the skill bot. The skill bot designer specifies a dialog flow using a language, such as markdown language. In certain embodiments, a version of YAML called OBotML may be used to specify a dialog flow for a skill bot. The dialog flow definition for a skill bot acts as a model for the conversation itself, one that lets the skill bot designer choreograph the interactions between a skill bot and the users that the skill bot services.

In certain embodiments, the dialog flow definition for a skill bot contains three sections:

(a) a context section

(b) a default transitions section

(c) a states section

Context section—The skill bot designer can define variables that are used in a conversation flow in the context section. Other variables that may be named in the context section include, without limitation: variables for error handling, variables for built-in or custom entities, user variables that enable the skill bot to recognize and persist user preferences, and the like.

Default transitions section—Transitions for a skill bot can be defined in the dialog flow states section or in the default transitions section. The transitions defined in the default transition section act as a fallback and get triggered when there are no applicable transitions defined within a state, or the conditions required to trigger a state transition cannot be met. The default transitions section can be used to define routing that allows the skill bot to gracefully handle unexpected user actions.

States section—A dialog flow and its related operations are defined as a sequence of transitory states, which manage the logic within the dialog flow. Each state node within a dialog flow definition names a component that provides the functionality needed at that point in the dialog. States are thus built around the components. A state contains component-specific properties and defines the transitions to other states that get triggered after the component executes.

Special case scenarios may be handled using the states sections. For example, there might be times when you want to provide users the option to temporarily leave a first skill they are engaged with to do something in a second skill within the digital assistant. For example, if a user is engaged in a conversation with a shopping skill (e.g., the user has made some selections for purchase), the user may want to jump to a banking skill (e.g., the user may want to ensure that he/she has enough money for the purchase), and then return to the shopping skill to complete the user's order. To address this, an action in the first skill can be configured to initiate an interaction with the second different skill in the same digital assistant and then return to the original flow.

(6) Adding custom components to the skill bot—As described above, states specified in a dialog flow for a skill bot name components that provide the functionality needed corresponding to the states. Components enable a skill bot to perform functions. In certain embodiments, DABP **102** provides a set of preconfigured components for performing a wide range of functions. A skill bot designer can select one of more of these preconfigured components and associate them with states in the dialog flow for a skill bot. The skill bot designer can also create custom or new components using tools provided by DABP **102** and associate the custom components with one or more states in the dialog flow for a skill bot.

(7) Testing and deploying the skill bot—DABP **102** provides several features that enable the skill bot designer to test a skill bot being developed. The skill bot can then be deployed and included in a digital assistant.

While the description above describes how to create a skill bot, similar techniques may also be used to create a digital assistant (or the master bot). At the master bot or digital assistant level, built-in system intents may be configured for the digital assistant. These built-in system intents are used to identify general tasks that the digital assistant itself (i.e., the master bot) can handle without invoking a skill bot associated with the digital assistant. Examples of system intents defined for a master bot include: (1) Exit: applies when the user signals the desire to exit the current conversation or context in the digital assistant; (2) Help: applies when the user asks for help or orientation; and (3) UnresolvedIntent: applies to user input that doesn't match well with the exit and help intents. The digital assistant also stores information about the one or more skill bots associated with the digital assistant. This information enables the master bot to select a particular skill bot for handling an utterance.

At the master bot or digital assistant level, when a user inputs a phrase or utterance to the digital assistant, the digital assistant is configured to perform processing to determine how to route the utterance and the related conversation. The digital assistant determines this using a routing model, which can be rules-based, AI-based, or a combination thereof. The digital assistant uses the routing model to determine whether the conversation corresponding to the user input utterance is to be routed to a particular skill for handling, is to be handled by the digital assistant or master bot itself per a built-in system intent, or is to be handled as a different state in a current conversation flow.

In certain embodiments, as part of this processing, the digital assistant determines if the user input utterance explicitly identifies a skill bot using its invocation name. If an invocation name is present in the user input, then it is treated as explicit invocation of the skill bot corresponding to the invocation name. In such a scenario, the digital assistant may route the user input to the explicitly invoked skill bot for further handling. If there is no specific or explicit invocation, in certain embodiments, the digital assistant evaluates the received user input utterance and computes confidence scores for the system intents and the skill bots associated with the digital assistant. The score computed for a skill bot or system intent represents how likely the user input is representative of a task that the skill bot is configured to perform or is representative of a system intent. Any system intent or skill bot with an associated computed confidence score exceeding a threshold value (e.g., a Confidence Threshold routing parameter) is selected as a candidate for further evaluation. The digital assistant then selects, from the identified candidates, a particular system intent or a skill bot for further handling of the user input utterance. In certain embodiments, after one or more skill bots are identified as candidates, the intents associated with those candidate skills are evaluated (according to the intent model for each skill) and confidence scores are determined for each intent. In general, any intent that has a confidence score exceeding a threshold value (e.g., 70%) is treated as a candidate intent. If a particular skill bot is selected, then the user utterance is routed to that skill bot for further processing. If a system intent is selected, then one or more actions are performed by the master bot itself according to the selected system intent.

FIG. **2** is a simplified block diagram of a master bot (MB) system **200** according to certain embodiments. MB system **200** can be implemented in software only, hardware only, or a combination of hardware and software. MB system **200** includes a pre-processing subsystem **210**, a multiple intent subsystem (MIS) **220**, an explicit invocation subsystem (EIS) **230**, a skill bot invoker **240**, and a data store **250**. MB system **200** depicted in FIG. **2** is merely an example of an arrangement of components in a master bot. One of ordinary skill in the art would recognize many possible variations, alternatives, and modifications. For example, in some implementations, MB system **200** may have more or fewer systems or components than those shown in FIG. **2**, may combine two or more subsystems, or may have a different configuration or arrangement of subsystems.

Pre-processing subsystem **210** receives an utterance "A" **202** from a user and processes the utterance through a language detector **212** and a language parser **214**. As indicated above, an utterance can be provided in various ways including audio or text. The utterance **202** can be a sentence fragment, a complete sentence, multiple sentences, and the like. Utterance **202** can include punctuation. For example, if the utterance **202** is provided as audio, the pre-processing subsystem **210** may convert the audio to text using a speech-to-text converter (not shown) that inserts punctuation marks into the resulting text, e.g., commas, semicolons, periods, etc.

Language detector **212** detects the language of the utterance **202** based on the text of the utterance **202**. The manner in which the utterance **202** is handled depends on the language since each language has its own grammar and semantics. Differences between languages are taken into consideration when analyzing the syntax and structure of an utterance.

Language parser **214** parses the utterance **202** to extract part of speech (POS) tags for individual linguistic units (e.g., words) in the utterance **202**. POS tags include, for example, noun (NN), pronoun (PN), verb (VB), and the like. Language parser **214** may also tokenize the linguistic units of the utterance **202** (e.g., to convert each word into a separate token) and lemmatize words. A lemma is the main form of a set of words as represented in a dictionary (e.g., "run" is the lemma for run, runs, ran, running, etc.). Other types of pre-processing that the language parser **214** can perform include chunking of compound expressions, e.g., combining "credit" and "card" into a single expression "credit card." Language parser **214** may also identify relationships between the words in the utterance **202**. For example, in some embodiments, the language parser **214** generates a dependency tree that indicates which part of the utterance (e.g. a particular noun) is a direct object, which part of the utterance is a preposition, and so on. The results of the processing performed by the language parser **214** form extracted information **205** and are provided as input to MIS **220** together with the utterance **202** itself.

As indicated above, the utterance **202** can include more than one sentence. For purposes of detecting multiple intents and explicit invocation, the utterance **202** can be treated as a single unit even if it includes multiple sentences. However, in certain embodiments, pre-processing can be performed, e.g., by the pre-processing subsystem **210**, to identify a single sentence among multiple sentences for multiple intents analysis and explicit invocation analysis. In general, the results produced by MIS **220** and EIS **230** are substantially the same regardless of whether the utterance **202** is processed at the level of an individual sentence or as a single unit comprising multiple sentences.

MIS **220** determines whether the utterance **202** represents multiple intents. Although MIS **220** can detect the presence of multiple intents in the utterance **202**, the processing performed by MIS **220** does not involve determining whether the intents of the utterance **202** match to any intents that have been configured for a bot. Instead, processing to determine whether an intent of the utterance **202** matches a bot intent can be performed by an intent classifier **242** of the MB system **200** or by an intent classifier of a skill bot (e.g., as shown in the embodiment of FIG. **3**). The processing performed by MIS **220** assumes that there exists a bot (e.g., a particular skill bot or the master bot itself) that can handle the utterance **202**. Therefore, the processing performed by MIS **220** does not require knowledge of what bots are in the chatbot system (e.g., the identities of skill bots registered with the master bot) or knowledge of what intents have been configured for a particular bot.

To determine that the utterance **202** includes multiple intents, the MIS **220** applies one or more rules from a set of rules **252** in the data store **250**. The rules applied to the utterance **202** depend on the language of the utterance **202** and may include sentence patterns that indicate the presence of multiple intents. For example, a sentence pattern may include a coordinating conjunction that joins two parts (e.g., conjuncts) of a sentence, where both parts correspond to a separate intent. If the utterance **202** matches the sentence pattern, it can be inferred that the utterance **202** represents multiple intents. It should be noted that an utterance with multiple intents does not necessarily have different intents (e.g., intents directed to different bots or to different intents within the same bot). Instead, the utterance could have separate instances of the same intent, e.g. "Place a pizza order using payment account X, then place a pizza order using payment account Y."

As part of determining that the utterance **202** represents multiple intents, the MIS **220** also determines what portions of the utterance **202** are associated with each intent. MIS **220** constructs, for each intent represented in an utterance containing multiple intents, a new utterance for separate processing in place of the original utterance, e.g., an utterance "B" **206** and an utterance "C" **208**, as depicted in FIG. **2**. Thus, the original utterance **202** can be split into two or more separate utterances that are handled one at a time. MIS **220** determines, using the extracted information **205** and/or from analysis of the utterance **202** itself, which of the two or more utterances should be handled first. For example, MIS **220** may determine that the utterance **202** contains a marker word indicating that a particular intent should be handled first. The newly formed utterance corresponding to this particular intent (e.g., one of utterance **206** or utterance **208**) will be the first to be sent for further processing by EIS **230**. After a conversation triggered by the first utterance has ended (or has been temporarily suspended), the next highest priority utterance (e.g., the other one of utterance **206** or utterance **208**) can then be sent to the EIS **230** for processing.

EIS **230** determines whether the utterance that it receives (e.g., utterance **206** or utterance **208**) contains an invocation name of a skill bot. In certain embodiments, each skill bot in a chatbot system is assigned a unique invocation name that distinguishes the skill bot from other skill bots in the chatbot system. A list of invocation names can be maintained as part of skill bot information **254** in data store **250**. An utterance is deemed to be an explicit invocation when the utterance contains a word match to an invocation name. If a bot is not explicitly invoked, then the utterance received by the EIS **230** is deemed a non-explicitly invoking utterance **234** and is input to an intent classifier (e.g., intent classifier **242**) of the master bot to determine which bot to use for handling the utterance. In some instances, the intent classifier **242** will determine that the master bot should handle a non-explicitly invoking utterance. In other instances, the intent classifier **242** will determine a skill bot to route the utterance to for handling.

The explicit invocation functionality provided by the EIS **230** has several advantages. It can reduce the amount of processing that the master bot has to perform. For example, when there is an explicit invocation, the master bot may not have to do any intent classification analysis (e.g., using the intent classifier **242**), or may have to do reduced intent classification analysis for selecting a skill bot. Thus, explicit invocation analysis may enable selection of a particular skill bot without resorting to intent classification analysis.

Also, there may be situations where there is an overlap in functionalities between multiple skill bots. This may happen, for example, if the intents handled by the two skill bots overlap or are very close to each other. In such a situation, it may be difficult for the master bot to identify which of the multiple skill bots to select based upon intent classification analysis alone. In such scenarios, the explicit invocation disambiguates the particular skill bot to be used.

In addition to determining that an utterance is an explicit invocation, the EIS **230** is responsible for determining whether any portion of the utterance should be used as input to the skill bot being explicitly invoked. In particular, EIS **230** can determine whether part of the utterance is not associated with the invocation. The EIS **230** can perform this determination through analysis of the utterance and/or analysis of the extracted information **205**. EIS **230** can send the part of the utterance not associated with the invocation to the invoked skill bot in lieu of sending the entire utterance that was received by the EIS **230**. In some instances, the input to the invoked skill bot is formed simply by removing any portion of the utterance associated with the invocation. For example, "I want to order pizza using Pizza Bot" can be shortened to "I want to order pizza" since "using Pizza Bot" is relevant to the invocation of the pizza bot, but irrelevant to any processing to be performed by the pizza bot. In some instances, EIS **230** may reformat the part to be sent to the invoked bot, e.g., to form a complete sentence. Thus, the EIS **230** determines not only that there is an explicit invocation, but also what to send to the skill bot when there is an explicit invocation. In some instances, there may not be any text to input to the bot being invoked. For example, if the utterance was "Pizza Bot", then the EIS **230** could determine that the pizza bot is being invoked, but there is no text to be processed by the pizza bot. In such scenarios, the EIS **230** may indicate to the skill bot invoker **240** that there is nothing to send.

Skill bot invoker **240** invokes a skill bot in various ways. For instance, skill bot invoker **240** can invoke a bot in response to receiving an indication **235** that a particular skill bot has been selected as a result of an explicit invocation. The indication **235** can be sent by the EIS **230** together with the input for the explicitly invoked skill bot. In this scenario, the skill bot invoker **240** will turn control of the conversation over to the explicitly invoked skill bot. The explicitly invoked skill bot will determine an appropriate response to the input from the EIS **230** by treating the input as a stand-alone utterance. For example, the response could be to perform a specific action or to start a new conversation in a particular state, where the initial state of the new conversation depends on the input sent from the EIS **230**.

Another way in which skill bot invoker **240** can invoke a skill bot is through implicit invocation using the intent classifier **242**. The intent classifier **242** can be trained, using machine-learning and/or rules-based training techniques, to determine a likelihood that an utterance is representative of a task that a particular skill bot is configured to perform. The intent classifier **242** is trained on different classes, one class for each skill bot. For instance, whenever a new skill bot is registered with the master bot, a list of example utterances associated with the new skill bot can be used to train the intent classifier **242** to determine a likelihood that a particular utterance is representative of a task that the new skill bot can perform. The parameters produced as result of this training (e.g., a set of values for parameters of a machine-learning model) can be stored as part of skill bot information **254**.

In certain embodiments, the intent classifier **242** is implemented using a machine-learning model, as described in further detail herein. Training of the machine-learning model may involve inputting at least a subset of utterances from the example utterances associated with various skill bots to generate, as an output of the machine-learning model, inferences as to which bot is the correct bot for handling any particular training utterance. For each training utterance, an indication of the correct bot to use for the training utterance may be provided as ground truth information. The behavior of the machine-learning model can then be adapted (e.g., through back-propagation) to minimize the difference between the generated inferences and the ground truth information.

In certain embodiments, the intent classifier **242** determines, for each skill bot registered with the master bot, a confidence score indicating a likelihood that the skill bot can handle an utterance (e.g., the non-explicitly invoking utterance **234** received from EIS **230**). The intent classifier **242** may also determine a confidence score for each system level intent (e.g., help, exit) that has been configured. If a particular confidence score meets one or more conditions, then the skill bot invoker **240** will invoke the bot associated with the particular confidence score. For example, a threshold confidence score value may need to be met. Thus, an output **245** of the intent classifier **242** is either an identification of a system intent or an identification of a particular skill bot. In some embodiments, in addition to meeting a threshold confidence score value, the confidence score must exceed the next highest confidence score by a certain win margin. Imposing such a condition would enable routing to a particular skill bot when the confidence scores of multiple skill bots each exceed the threshold confidence score value.

After identifying a bot based on evaluation of confidence scores, the skill bot invoker **240** hands over processing to the identified bot. In the case of a system intent, the identified bot is the master bot. Otherwise, the identified bot is a skill bot. Further, the skill bot invoker **240** will determine what to provide as input **247** for the identified bot. As indicated above, in the case of an explicit invocation, the input **247** can be based on a part of an utterance that is not associated with the invocation, or the input **247** can be nothing (e.g., an empty string). In the case of an implicit invocation, the input **247** can be the entire utterance.

Data store **250** comprises one or more computing devices that store data used by the various subsystems of the master bot system **200**. As explained above, the data store **250** includes rules **252** and skill bot information **254**. The rules **252** include, for example, rules for determining, by MIS **220**, when an utterance represents multiple intents and how to split an utterance that represents multiple intents. The rules **252** further include rules for determining, by EIS **230**, which parts of an utterance that explicitly invokes a skill bot to send to the skill bot. The skill bot information **254** includes invocation names of skill bots in the chatbot system, e.g., a list of the invocation names of all skill bots registered with a particular master bot. The skill bot information **254** can also include information used by intent classifier **242** to determine a confidence score for each skill bot in the chatbot system, e.g., parameters of a machine-learning model.

FIG. **3** is a simplified block diagram of a skill bot system **300** according to certain embodiments. Skill bot system **300** is a computing system that can be implemented in software only, hardware only, or a combination of hardware and software. In certain embodiments such as the embodiment depicted in FIG. **1**, skill bot system **300** can be used to implement one or more skill bots within a digital assistant.

Skill bot system **300** includes an MIS **310**, an intent classifier **320**, and a conversation manager **330**. The MIS **310** is analogous to the MIS **220** in FIG. **2** and provides similar functionality, including being operable to determine, using rules **352** in a data store **350**: (1) whether an utterance represents multiple intents and, if so, (2) how to split the utterance into a separate utterance for each intent of the multiple intents. In certain embodiments, the rules applied by MIS **310** for detecting multiple intents and for splitting an utterance are the same as those applied by MIS **220**. The MIS **310** receives an utterance **302** and extracted information **304**. The extracted information **304** is analogous to the extracted information **205** in FIG. **1** and can be generated using the language parser **214** or a language parser local to the skill bot system **300**.

Intent classifier **320** can be trained in a similar manner to the intent classifier **242** discussed above in connection with the embodiment of FIG. **2** and as described in further detail herein. For instance, in certain embodiments, the intent classifier **320** is implemented using a machine-learning model. The machine-learning model of the intent classifier **320** is trained for a particular skill bot, using at least a subset of example utterances associated with that particular skill bot as training utterances. The ground truth for each training utterance would be the particular bot intent associated with the training utterance.

The utterance **302** can be received directly from the user or supplied through a master bot. When the utterance **302** is supplied through a master bot, e.g., as a result of processing through MIS **220** and EIS **230** in the embodiment depicted in FIG. **2**, the MIS **310** can be bypassed so as to avoid repeating processing already performed by MIS **220**. However, if the utterance **302** is received directly from the user, e.g., during a conversation that occurs after routing to a skill bot, then MIS **310** can process the utterance **302** to determine whether the utterance **302** represents multiple intents. If so, then MIS **310** applies one or more rules to split the utterance **302** into a separate utterance for each intent, e.g., an utterance "D" **306** and an utterance "E" **308**. If utterance **302** does not represent multiple intents, then MIS **310** forwards the utterance **302** to intent classifier **320** for intent classification and without splitting the utterance **302**.

Intent classifier **320** is configured to match a received utterance (e.g., utterance **306** or **308**) to an intent associated with skill bot system **300**. As explained above, a skill bot can be configured with one or more intents, each intent including at least one example utterance that is associated with the intent and used for training a classifier. In the embodiment of FIG. **2**, the intent classifier **242** of the master bot system **200** is trained to determine confidence scores for individual skill bots and confidence scores for system intents. Similarly, intent classifier **320** can be trained to determine a confidence score for each intent associated with the skill bot system **300**. Whereas the classification performed by intent classifier **242** is at the bot level, the classification performed by intent classifier **320** is at the intent level and therefore finer grained. The intent classifier **320** has access to intents information **354**. The intents information **354** includes, for each intent associated with the skill bot system **300**, a list of utterances that are representative of and illustrate the meaning of the intent and are typically associated with a task performable by that intent. The intents information **354** can further include parameters produced as a result of training on this list of utterances.

Conversation manager **330** receives, as an output of intent classifier **320**, an indication **322** of a particular intent, identified by the intent classifier **320**, as best matching the utterance that was input to the intent classifier **320**. In some instances, the intent classifier **320** is unable to determine any match. For example, the confidence scores computed by the intent classifier **320** could fall below a threshold confidence score value if the utterance is directed to a system intent or an intent of a different skill bot. When this occurs, the skill bot system **300** may refer the utterance to the master bot for handling, e.g., to route to a different skill bot. However, if the intent classifier **320** is successful in identifying an intent within the skill bot, then the conversation manager **330** will initiate a conversation with the user.

The conversation initiated by the conversation manager **330** is a conversation specific to the intent identified by the intent classifier **320**. For instance, the conversation manager **330** may be implemented using a state machine configured to execute a dialog flow for the identified intent. The state machine can include a default starting state (e.g., for when the intent is invoked without any additional input) and one or more additional states, where each state has associated with it actions to be performed by the skill bot (e.g., executing a purchase transaction) and/or dialog (e.g., questions, responses) to be presented to the user. Thus, the conversation manager **330** can determine an action/dialog **335** upon receiving the indication **322** identifying the intent, and can determine additional actions or dialog in response to subsequent utterances received during the conversation.

Data store **350** comprises one or more computing devices that store data used by the various subsystems of the skill bot system **300**. As depicted in FIG. **3**, the data store **350** includes the rules **352** and the intents information **354**. In certain embodiments, data store **350** can be integrated into a data store of a master bot or digital assistant, e.g., the data store **250** in FIG. **2**.

Chatbot Training And Deployment System

In some embodiments, a machine-learning model is trained using path dropout to improve runtime performance of language processing tasks such as named entity recognition. The machine-learning model may be implemented in a chatbot system, as described with respect to FIGS. **1**, **2** and **3**.

FIG. **4**A is a simplified block diagram of a chatbot training and deployment system **400** in accordance with various embodiments. The chatbot training and deployment system **400** takes as input an input utterance **402**, which may be text or spoken speech corresponding to an utterance. The input utterance **402** is processed by a tokenizer **404** to produce tokens **406**. The tokens **406** are units of text, which may include words, parts of words, characters, or sets of words that are identified in the input utterance **402** by the tokenizer **404** (e.g., I-want-to-go-to-Jamaica, where the tokens are separated by dashes).

The tokens **406** are provided as input to an encoder **408**. The encoder **408** may, for example, be part of a pretrained language model (e.g., Bidirectional Encoder Representations from Transformers (BERT)) or another suitable encoder. BERT is a transformer-based network that includes an encoder component and a decoder component. (See Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," arXiv: 1810.04805 (2018)). As a specific example, the encoder **408** is the bottom n (8) layers of BERT. In some examples, the encoder includes 1 embedding layer and n−1 (7) transformer layers. The encoder **408** generates feature embeddings **409**. The feature embeddings **409** can be dense vector representations of tokens (e.g., subwords, words, etc.) in lower dimensional space. The feature embeddings **409** are provided as input to components of a machine-learning model **410**, such as that depicted in FIG. **4**B.

Machine-Learning Model with Dropout Path

FIG. **4**B shows a block diagram illustrating components of a machine-learning model **410** for performing language processing tasks, including a dropout path corresponding to a second connection **412**. The components of the machine-learning model **410** may be part of a larger model or set of models, as described above with respect to FIG. **4**A and FIGS. **1-3**.

The machine-learning model **410** can be, or be part of, a named entity recognizer model configured and used to (for example) recognize one or more entities in an utterance. In some aspects, the machine-learning model **410** is, or is part of, a pretrained language model such as BERT. In some examples, components shown in **410** correspond to the BERT decoder.

In some embodiments, the machine-learning model **410** includes a transformer encoder block or several transformer encoder blocks. The first main component of a transformer encoder block is a multi-head self-attention component, which concatenates the outputs of multiple self-attention networks. (See Vaswani et al., "Attention is All You Need,"

31^{st} *Conf. on Neural Info. Processing Systs.* (NPIS 2017), FIG. **2**). A residual connection and normalization are applied to the outputs of the multi-head self-attention component, which is then fed into a Feed-Forward Neural Network followed by a residual connection and normalization step.

The machine-learning model **410** can include multiple layers, multiple layers that use a multi-head self-attention technique, and/or multiple layers that include one or more transformer blocks.

The machine learning model **410** includes a multi-head self-attention network **414**. The multi-head self-attention network is the concatenation of the outputs of multiple layers and multiple attention heads. (See Vaswani, supra). Each attention head may provide an output that is concatenated.

The machine learning model **410** includes a first connection **411**. The first connection **411** provides input embeddings **409** (an encoded representation of an input utterance **402**) as input to the multi-head self-attention network **414**. The input embeddings **409** are provided as inputs to the multi-head self-attention network **414** via the first connection **411**.

For a multi-head self-attention technique, each of the multiple heads performs a self-attention technique using different Query, Key, and Value linear projections. A self-attention technique can transform the given input embeddings into each of a Query representation (generated by using a Query linear projection), a Key representation (generated by using a Key linear projection), and a Value representation (generated by using a Value linear projection). Multiplying the Query representation by the Key representation (and potentially applying an activation function, such as a softmax function) to generate attention scores that indicate how much attention is to be paid to a value at a given position when evaluating a value at another position. The attention scores can then be multiplied by the Value representation to generate an output. This can be done using an attention weight matrix to organize the attention scores in a matrix to exploit the power of parallelization in batch processing. The output of the multi-head self-attention network **414** may include information associated with dependencies for each token, such as syntactic or semantic relationships between tokens. (See Vaswani et al., supra).

The output of the multi-head self-attention network **414** is provided to normalization component **416**. The normalization component **416** may normalize received input. For example, the normalization component computes and outputs a layer normalization based on input x and Sublayer(x), the function implemented by the sub-layer.

The machine learning model **410** further includes a second connection **412** that passes the input embeddings **409** to the normalization component **416** directly, bypassing the multi-head self-attention network **414**. In some implementations, the second connection **412** is a residual connection, which is a path for data to reach latter parts of the machine-learning model by skipping one or more layers.

In some embodiments, the second connection **412** is dropped out (as indicated by the "X" **413**). The second connection **412** can be dropped out, for example, by including a dropout layer before or after the second connection **412**. By dropping out the second connection **412**, the input embeddings will be increasingly processed by the multi-head self-attention network **414**.

In some embodiments, the machine learning model **410** further includes a feed forward component **418** and a second normalization component **420** for adding and normalizing the output of the feed forward component **418**. The machine learning model **410** may include a third path through the feed forward component **418** and a fourth path connecting the output of the first normalization component **416** to the second normalization component **420**.

The ultimate output **422** of the machine-learning model may, for example, be a named entity type corresponding to one or more words identified from the input. For example, the output corresponding to the utterance "What is the amount I spent at Joe's Shoe Depot" identifies Joe's Shoe Depot as a named entity of type MERCHANT_NAME. In some aspects, multiple machine-learning models **410** are included in the chatbot training and deployment system **400**. The output **422** may be provided to a next machine-learning model, and so forth, for a set of machine-learning models such as transformer blocks.

At the chatbot implementation stage, the machine-learning model **410** may be used in combination with one or more other models, such as another model for determining a likelihood that an utterance is representative of a task that a particular skill bot is configured to perform, another model for predicting an intent from an utterance for a first type of skill bot, and/or another model for predicting an intent from an utterance for a second type of skill bot.

Techniques for Generating and Using a Machine-Learning Model with Path Dropout

FIG. **5**A is a flowchart illustrating a process **500** for generating and using a machine learning model with path dropout according to certain embodiments. The processing depicted in FIG. **5**A may be implemented in software (e.g., code, instructions, program) executed by one or more processing units (e.g., processors, cores) of the respective systems, hardware, or combinations thereof. The software may be stored on a non-transitory storage medium (e.g., on a memory device). The method presented in FIG. **5**A and described below is intended to be illustrative and non-limiting. Although FIG. **5**A depicts the various processing steps occurring in a particular sequence or order, this is not intended to be limiting. In certain alternative embodiments, the steps may be performed in some different order or some steps may also be performed in parallel.

At block **502**, a machine-learning model is accessed. The machine-learning model includes one or more blocks. In some embodiments, the blocks are transformer blocks, which process input data using self-attention, as described above with respect to FIG. **4**B. Each block includes a multi-head self-attention network. As described above with respect to FIG. **4**B, each multi-head self-attention network includes multiple layers configured to implement a multi-head self-attention technique. The machine-learning model also includes a first connection for providing input to the multi-head self-attention network (e.g., first connection **411** shown in FIG. **4**B). The machine-learning model also includes a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network (e.g., second connection **412** shown in FIG. **4**B).

At block **504**, a training data set is accessed. The training data set includes pairs of input data (e.g., sample utterances) and corresponding labels (e.g., a named entity in the utterance).

The training data set may include example utterances associated with one or more skill bots. As indicated above, an utterance is in textual format. The utterance can be a sentence fragment, a complete sentence, multiple sentences, and the like. In some instances, the example utterances are provided by a previous or existing client or customer. In other instances, the example utterances are automatically generated from prior libraries of utterances (e.g., identifying utterances from a library that are specific to a skill that a chatbot is designated to learn).

The training data may also include training labels, where each of the training labels corresponds to an individual training utterance. The training labels may be stored in or stored as a matrix or table of values. For each training utterance, an indication of the correct entities and classification thereof to be inferred may be provided as ground truth information for training labels. The behavior of the model being trained can then be adapted (e.g., through back-propagation) to minimize the difference between the generated inferences for various entities and the ground truth information.

At block **506**, a dropout parameter is identified. In some implementations, the dropout parameter is a dropout rate. The dropout parameter establishes a likelihood that something (e.g., the second connection **412** of FIG. **4**, or attention weights, as described below) will be dropped out (i.e., blocked or ignored) on a particular training iteration. For example, if the second connection has a dropout parameter of 0, it will not be dropped out at all, if the second has a dropout parameter of 0.31, then the second connection will be dropped out in some of the training iterations, and if the second connection has a dropout parameter of 0.9, it will be dropped out in many (i.e., a higher percentage of) of the training iterations.

In some embodiments, the dropout parameter is a hyperparameter of the machine-learning model. The hyperparameters are settings that can be tuned or optimized to control the behavior of the machine learning model.

The dropout parameter may be identified by performing a hypertuning process. The system may perform a hyperparameter optimization process to select an appropriate value for the dropout parameter. In some implementations, the dropout parameter is identified using suitable hyperparameter optimization techniques such as grid search, random search, or Bayesian optimization. The hyperparameters can be established using a hyperparameter tuning algorithm which chooses values for these hyperparameters to optimize or minimize an objective function which reflects the model's performance. The model's performance can be measured on validation or test datasets that are separate from the training datasets. In an example of a suitable hypertuning process, users first define a search space (set of possible values) for each hyperparameter. There are multiple processes in which each will run a fixed set of N trials, where N is configurable. The findings are saved to a shared database. For each trial, a process will select a set of hyperparameters (using past trials or random search, etc.) and then perform full end-to-end training and evaluation. The trial score is computed using the evaluation result. After trials are executed, the trial with the best score is selected.

At block **508**, the dropout parameter is applied to the second connection. The second connection is dropped out according to the dropout parameter identified at block **506**. In some embodiments, the second connection is a residual connection and the dropout parameter is a dropout rate. Applying the dropout parameter includes dropping out the residual connection according to the dropout rate. Based on the dropout parameter, the second connection is not used and input flows through the first connection. This may happen on a subset of training iterations according to the dropout rate.

Alternatively, or additionally, dropout is applied to an attention weight matrix. The attention weight matrix is made up of scores for pairs of tokens: the token at the position that the model is trying to compute the contextual representation of and the surrounding tokens (including itself). The tokens can be, e.g., words or other units of text or spoken speech. For example, input representation feature maps for a pair of sentences s_0 and s_1 are matched to arrive at the attention weight matrix A, where each cell in the attention matrix, A_{ij}, represents the attention score between the i^{th} token in so and j^{th} token in s_1. (See Mahendran Venkatachalam, "Attention in Neural Networks," *Towards Data Science*, available at https://towardsdatascience.com/attention-in-neural-networks-e66920838742 (2019)) (i^{th} token and j^{th} token can belong to the same sentence, hence "self" in "self-attention"). The attention weight matrix may contain attention weights from each token to each other token, including itself. For example, for 100 tokens, the attention weight matrix will be of size 100×100. Diagonal entries in the attention weight matrix represent the attention from a token to itself. If these diagonal entries are dropped, this further forces to model to pay less attention to the token itself and more attention to the surrounding tokens. FIG. **5**B illustrates an example of pairings of tokens, as in an attention weight matrix. In this example, dropout is applied to the diagonal entries, so that tokens such as "evergreen" will have zero attention score with themselves, as described in further detail below with respect to FIG. **5**B.

In some aspects, a first dropout parameter is identified for the second connection, and a second dropout parameter is identified for the attention weight matrix. Depending on the dropout parameter(s) (e.g., selected by the hypertuner as described above), the second connection and/or entries in the attention weight matrix may be dropped out on a particular training iteration. For example, on one training iteration, both the second connection and the diagonal entries in the attention weight matrix are dropped out; on a second training iteration, the second connection is dropped out and the diagonal entries in the attention weight matrix are not dropped out; on a third training iteration, the diagonal entries in the attention weight matrix are dropped out and the second connection is not dropped out; and so forth.

At block **510**, the machine-learning model is trained using the training data set to generate a trained machine-learning model. During training, the second connection and/or the diagonal entries in the attention weight matrix are dropped out according to the dropout parameter identified at block **506** and applied at block **508**.

The training can include processing each of one or more training data elements (e.g., input utterances or corresponding encodings) using a current version of the model to generate a prediction, calculating a loss based on the prediction and a label corresponding to the data element, and updating one or more parameters of the model based on the loss. In order to generate the prediction, multiple intermediate values may be generated for each training data element. For example, an intermediate value may include an output from a layer in the model. Each layer may generate multiple outputs (e.g., over 100, over 1,000, over 10,000, or over 100,000 outputs). In some instances, the model in its entirety is configured to generate a consistent output so long as an input is the same, the model's hyperparameters are the same, and the model's parameters are the same. Once training is complete, a trained machine-learning model is the result, which can be or can be part of an entity recognition model.

By applying dropout (e.g., when the second connection and/or the entries in the attention weight matrix are dropped out), in the training process, the machine-learning model is caused to learn contextual information through multi-headed self-attention. As noted above, the training data set may include one or more utterances. The utterances can include one or more words that are labeled as an entity (e.g., California, Record Depot, etc.). The utterances can also include one or more words that are not labeled as an entity (e.g., in, the, how, get, fast, etc.). Those words that are not labeled as entities can be used to determine the contextual information. For example, in the utterance "I need directions to Sydney," the word Sydney will be labeled as an entity (city) and the remaining words (I, need, directions, to) provide the context that Sydney is a geographic location. This sort of contextual information can help the model more accurately identify entities, particularly if there is any ambiguity that the context can resolve.

At block **512**, use of the trained machine-learning model is facilitated. For example, the trained machine-learning model is provided an input utterance and identifies an entity in the input utterance. Alternatively, or additionally, the trained machine learning model can be used to identify an intent associated with the utterance and/or to identify a response for the utterance. In some examples, facilitating use of the trained machine-learning model includes inputting an utterance to the trained machine-learning model to identify an entity. The trained machine-learning model processes the input utterance to identify one or more entities therein. The system executing the machine-learning model may then output the entity or perform further processing, such as determining an intent, determining a response, or the like. The system executing the machine-learning model, or a downstream system receiving the identified entity or a derivative thereof, may then prepare and provide a response to the utterance based on the identified entity. For example, the response can be audio output answering a question, displayed text, displayed images, or the like.

At implementation, the trained machine-learning model may be deployed and used as part of or as a trained entity recognition model implementing one or more chatbots. For example, one or more chatbots may be configured with the trained machine-learning model to receive text data (e.g., corresponding to utterances) from one or more users, and recognize and extract entities from various utterances received by the one or more chatbots. Text data may include text data received in an environment or context corresponding to that for which the trained machine-learning model was trained. The entities may be part of extracted information obtained from the text data, and may be used in downstream processing such as intent classification. An output generated based on the extracted entities may then be transmitted to and/or presented at a device corresponding to a source of the utterance.

FIG. **5**B is a diagram **550** illustrating an example of token pairings with dropout, according to some embodiments. The example tokens are pay **552**, john **554**, $**556**, **10 558**, using **560**, evergreen **562**, and account **564**. In the attention weight matrix described above, each token will have some established degree of attention to each other token, including itself. Such pairings are indicated by the lines between the two columns.

As described above with respect to FIG. **5**A, in some embodiments, diagonal entries of an attention weight matrix are dropped out. The diagonal entries represent pairings of tokens to themselves. As shown in FIG. **5**B, when the diagonal entries in the attention weight matrix are dropped out, a token, such as evergreen **562**, retains attention to other tokens in an utterance (e.g., tokens **552**-**560** and **564** in this example), which may provide contextual information. The attention weight for pairings of tokens to themselves, such as evergreen **562** to evergreen **562**, are dropped or zeroed out, as indicated by the "X" **580**. When this dropout is applied in training, the machine-learning model is caused to learn to increasingly focus on contextual information, such as surrounding adjectives, verbs, and so forth.

Using the techniques described herein, by applying dropout to the second connection and/or the diagonal entries of the attention weight matrix, the machine-learning model is caused to learn contextual information through multi-headed self-attention. As a result, the model is forced to focus more on contextual information, which provides improved results. Table 2, below, illustrates example results with and without path dropout. The values shown are F1 scores, which are a weighted average of precision (the number of correct predictions divided by total prediction made by model) and recall (the number of true positives divided by the total number of true positives and false negatives). The F1 score conveys the balance between the precision and the recall, and higher F1 scores indicate better (e.g., more precise) results.

TABLE 2

EXAMPLE RESULTS WITH DROPPING OUT PATH

Config	CoNLL 2003	MIT-Movie	MIT-Restaurant	ATIS	Multi-WOZ	SNIPS	Wiki-Gold	Expense	PPM	Calendar	Average
Hypertuned + train_without_eval	0.892	0.717	0.787	0.957	0.885	0.943	0.769	0.867	0.993	0.825	0.8635
+regularization	0.893	0.718	0.793	0.955	0.888	0.946	0.780	0.880	0.993	0.849	0.8695
+dropping out path	0.893	0.725	0.793	0.959	0.889	0.948	0.772	0.885	0.993	0.984	0.8841

Illustrative Systems

FIG. **6** depicts a simplified diagram of a distributed system **600**. In the illustrated example, distributed system **600** includes one or more client computing devices **602**, **604**, **606**, and **608**, coupled to a server **612** via one or more communication networks **610**. Clients computing devices **602**, **604**, **606**, and **608** may be configured to execute one or more applications.

In various examples, server **612** may be adapted to run one or more services or software applications that enable one or more embodiments described in this disclosure. In certain examples, server **612** may also provide other services or software applications that may include non-virtual and virtual environments. In some examples, these services may be offered as web-based or cloud services, such as under a Software as a Service (SaaS) model to the users of client computing devices **602**, **604**, **606**, and/or **608**. Users operating client computing devices **602**, **604**, **606**, and/or **608** may in turn utilize one or more client applications to interact with server **612** to utilize the services provided by these components.

In the configuration depicted in FIG. **6**, server **612** may include one or more components **618**, **620** and **622** that implement the functions performed by server **612**. These components may include software components that may be executed by one or more processors, hardware components, or combinations thereof. It should be appreciated that various different system configurations are possible, which may be different from distributed system **600**. The example shown in FIG. **6** is thus one example of a distributed system for implementing an example system and is not intended to be limiting.

Users may use client computing devices **602**, **604**, **606**, and/or **608** to execute one or more applications, models or chatbots, which may generate one or more events or models that may then be implemented or serviced in accordance with the teachings of this disclosure. A client device may provide an interface that enables a user of the client device to interact with the client device. The client device may also output information to the user via this interface. Although FIG. **6** depicts only four client computing devices, any number of client computing devices may be supported.

The client devices may include various types of computing systems such as portable handheld devices, general purpose computers such as personal computers and laptops, workstation computers, wearable devices, gaming systems, thin clients, various messaging devices, sensors or other sensing devices, and the like. These computing devices may run various types and versions of software applications and operating systems (e.g., Microsoft Windows®, Apple Macintosh®, UNIX® or UNIX-like operating systems, Linux or Linux-like operating systems such as Google Chrome™ OS) including various mobile operating systems (e.g., Microsoft Windows Mobile®, iOS®, Windows Phone®, Android™, BlackBerry®, Palm OS®). Portable handheld devices may include cellular phones, smartphones, (e.g., an iPhone®), tablets (e.g., iPad®), personal digital assistants (PDAs), and the like. Wearable devices may include Google Glass® head mounted display, and other devices. Gaming systems may include various handheld gaming devices, Internet-enabled gaming devices (e.g., a Microsoft Xbox® gaming console with or without a Kinect® gesture input device, Sony PlayStation® system, various gaming systems provided by Nintendo®, and others), and the like. The client devices may be capable of executing various different applications such as various Internet-related apps, communication applications (e.g., E-mail applications, short message service (SMS) applications) and may use various communication protocols.

Network(s) **610** may be any type of network familiar to those skilled in the art that may support data communications using any of a variety of available protocols, including without limitation TCP/IP (transmission control protocol/Internet protocol), SNA (systems network architecture), IPX (Internet packet exchange), AppleTalk®, and the like. Merely by way of example, network(s) **610** may be a local area network (LAN), networks based on Ethernet, Token-Ring, a wide-area network (WAN), the Internet, a virtual network, a virtual private network (VPN), an intranet, an extranet, a public switched telephone network (PSTN), an infra-red network, a wireless network (e.g., a network operating under any of the Institute of Electrical and Electronics (IEEE) 1002.11 suite of protocols, Bluetooth®, and/or any other wireless protocol), and/or any combination of these and/or other networks.

Server **612** may be composed of one or more general purpose computers, specialized server computers (including, by way of example, PC (personal computer) servers, UNIX® servers, mid-range servers, mainframe computers, rack-mounted servers, etc.), server farms, server clusters, or any other appropriate arrangement and/or combination. Server **612** may include one or more virtual machines running virtual operating systems, or other computing architectures involving virtualization such as one or more flexible pools of logical storage devices that may be virtualized to maintain virtual storage devices for the server. In various examples, server **612** may be adapted to run one or more services or software applications that provide the functionality described in the foregoing disclosure.

The computing systems in server **612** may run one or more operating systems including any of those discussed above, as well as any commercially available server operating system. Server **612** may also run any of a variety of additional server applications and/or mid-tier applications, including HTTP (hypertext transport protocol) servers, FTP (file transfer protocol) servers, CGI (common gateway interface) servers, JAVA® servers, database servers, and the like. Exemplary database servers include without limitation those commercially available from Oracle®, Microsoft®, Sybase®, IBM® (International Business Machines), and the like.

In some implementations, server **612** may include one or more applications to analyze and consolidate data feeds and/or event updates received from users of client computing devices **602**, **604**, **606**, and **608**. As an example, data feeds and/or event updates may include, but are not limited to, Twitter® feeds, Facebook® updates or real-time updates received from one or more third party information sources and continuous data streams, which may include real-time events related to sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), clickstream analysis tools, automobile traffic monitoring, and the like. Server **612** may also include one or more applications to display the data feeds and/or real-time events via one or more display devices of client computing devices **602**, **604**, **606**, and **608**.

Distributed system **600** may also include one or more data repositories **614**, **616**. These data repositories may be used to store data and other information in certain examples. For example, one or more of the data repositories **614**, **616** may be used to store information such as information related to chatbot performance or generated models for use by chatbots used by server **612** when performing various functions in accordance with various embodiments. Data repositories **614**, **616** may reside in a variety of locations. For example, a data repository used by server **612** may be local to server **612** or may be remote from server **612** and in communication with server **612** via a network-based or dedicated connection. Data repositories **614**, **616** may be of different types. In certain examples, a data repository used by server **612** may be a database, for example, a relational database, such as databases provided by Oracle Corporation® and other vendors. One or more of these databases may be adapted to enable storage, update, and retrieval of data to and from the database in response to SQL-formatted commands.

In certain examples, one or more of data repositories **614**, **616** may also be used by applications to store application data. The data repositories used by applications may be of different types such as, for example, a key-value store repository, an object store repository, or a general storage repository supported by a file system.

In certain examples, the functionalities described in this disclosure may be offered as services via a cloud environment. FIG. **7** is a simplified block diagram of a cloud-based system environment in which various services may be offered as cloud services in accordance with certain examples. In the example depicted in FIG. **7**, cloud infrastructure system **702** may provide one or more cloud services that may be requested by users using one or more client computing devices **704**, **706**, and **708**. Cloud infrastructure system **702** may comprise one or more computers and/or servers that may include those described above for server **612**. The computers in cloud infrastructure system **702** may be organized as general purpose computers, specialized server computers, server farms, server clusters, or any other appropriate arrangement and/or combination.

Network(s) **710** may facilitate communication and exchange of data between clients **704**, **706**, and **708** and cloud infrastructure system **702**. Network(s) **710** may include one or more networks. The networks may be of the same or different types. Network(s) **710** may support one or more communication protocols, including wired and/or wireless protocols, for facilitating the communications.

The example depicted in FIG. **7** is only one example of a cloud infrastructure system and is not intended to be limiting. It should be appreciated that, in some other examples, cloud infrastructure system **702** may have more or fewer components than those depicted in FIG. **7**, may combine two or more components, or may have a different configuration or arrangement of components. For example, although FIG. **7** depicts three client computing devices, any number of client computing devices may be supported in alternative examples.

The term cloud service is generally used to refer to a service that is made available to users on demand and via a communication network such as the Internet by systems (e.g., cloud infrastructure system **702**) of a service provider. Typically, in a public cloud environment, servers and systems that make up the cloud service provider's system are different from the customer's own on-premise servers and systems. The cloud service provider's systems are managed by the cloud service provider. Customers may thus avail themselves of cloud services provided by a cloud service provider without having to purchase separate licenses, support, or hardware and software resources for the services. For example, a cloud service provider's system may host an application, and a user may, via the Internet, on demand, order and use the application without the user having to buy infrastructure resources for executing the application. Cloud services are designed to provide easy, scalable access to applications, resources and services. Several providers offer cloud services. For example, several cloud services are offered by Oracle Corporation® of Redwood Shores, California, such as middleware services, database services, Java cloud services, and others.

In certain examples, cloud infrastructure system **702** may provide one or more cloud services using different models such as under a Software as a Service (SaaS) model, a Platform as a Service (PaaS) model, an Infrastructure as a Service (IaaS) model, and others, including hybrid service models. Cloud infrastructure system **702** may include a suite of applications, middleware, databases, and other resources that enable provision of the various cloud services.

A SaaS model enables an application or software to be delivered to a customer over a communication network like the Internet, as a service, without the customer having to buy the hardware or software for the underlying application. For example, a SaaS model may be used to provide customers access to on-demand applications that are hosted by cloud infrastructure system **702**. Examples of SaaS services provided by Oracle Corporation® include, without limitation, various services for human resources/capital management, customer relationship management (CRM), enterprise resource planning (ERP), supply chain management (SCM), enterprise performance management (EPM), analytics services, social applications, and others.

An IaaS model is generally used to provide infrastructure resources (e.g., servers, storage, hardware and networking resources) to a customer as a cloud service to provide elastic compute and storage capabilities. Various IaaS services are provided by Oracle Corporation®.

A PaaS model is generally used to provide, as a service, platform and environment resources that enable customers to develop, run, and manage applications and services without the customer having to procure, build, or maintain such resources. Examples of PaaS services provided by Oracle Corporation® include, without limitation, Oracle Java Cloud Service (JCS), Oracle Database Cloud Service (DBCS), data management cloud service, various application development solutions services, and others.

Cloud services are generally provided on an on-demand self-service basis, subscription-based, elastically scalable, reliable, highly available, and secure manner. For example, a customer, via a subscription order, may order one or more services provided by cloud infrastructure system **702**. Cloud infrastructure system **702** then performs processing to provide the services requested in the customer's subscription order. For example, a user may use utterances to request the cloud infrastructure system to take a certain action (e.g., an intent), as described above, and/or provide services for a chatbot system as described herein. Cloud infrastructure system **702** may be configured to provide one or even multiple cloud services.

Cloud infrastructure system **702** may provide the cloud services via different deployment models. In a public cloud model, cloud infrastructure system **702** may be owned by a third party cloud services provider and the cloud services are offered to any general public customer, where the customer may be an individual or an enterprise. In certain other examples, under a private cloud model, cloud infrastructure system **702** may be operated within an organization (e.g., within an enterprise organization) and services provided to customers that are within the organization. For example, the customers may be various departments of an enterprise such as the Human Resources department, the Payroll department, etc. or even individuals within the enterprise. In certain other examples, under a community cloud model, the cloud infrastructure system **702** and the services provided may be shared by several organizations in a related community. Various other models such as hybrids of the above mentioned models may also be used.

Client computing devices **704**, **706**, and **708** may be of different types (such as client computing devices **602**, **604**, **606**, and **608** depicted in FIG. **6**) and may be capable of operating one or more client applications. A user may use a client device to interact with cloud infrastructure system **702**, such as to request a service provided by cloud infrastructure system **702**. For example, a user may use a client device to request information or action from a chatbot as described in this disclosure.

In some examples, the processing performed by cloud infrastructure system **702** for providing services may involve model training and deployment. This analysis may involve using, analyzing, and manipulating data sets to train and deploy one or more models. This analysis may be performed by one or more processors, possibly processing the data in parallel, performing simulations using the data, and the like. For example, big data analysis may be performed by cloud infrastructure system **702** for generating and training one or more models for a chatbot system. The data used for this analysis may include structured data (e.g., data stored in a database or structured according to a structured model) and/or unstructured data (e.g., data blobs (binary large objects)).

As depicted in the example in FIG. **7**, cloud infrastructure system **702** may include infrastructure resources **730** that are utilized for facilitating the provision of various cloud services offered by cloud infrastructure system **702**. Infrastructure resources **730** may include, for example, processing resources, storage or memory resources, networking resources, and the like. In certain examples, the storage virtual machines that are available for servicing storage requested from applications may be part of cloud infrastructure system **702**. In other examples, the storage virtual machines may be part of different systems.

In certain examples, to facilitate efficient provisioning of these resources for supporting the various cloud services provided by cloud infrastructure system **702** for different customers, the resources may be bundled into sets of resources or resource modules (also referred to as "pods"). Each resource module or pod may comprise a pre-integrated and optimized combination of resources of one or more types. In certain examples, different pods may be pre-provisioned for different types of cloud services. For example, a first set of pods may be provisioned for a database service, a second set of pods, which may include a different combination of resources than a pod in the first set of pods, may be provisioned for Java service, and the like. For some services, the resources allocated for provisioning the services may be shared between the services.

Cloud infrastructure system **702** may itself internally use services **732** that are shared by different components of cloud infrastructure system **702** and which facilitate the provisioning of services by cloud infrastructure system **702**. These internal shared services may include, without limitation, a security and identity service, an integration service, an enterprise repository service, an enterprise manager service, a virus scanning and white list service, a high availability, backup and recovery service, service for enabling cloud support, an email service, a notification service, a file transfer service, and the like.

Cloud infrastructure system **702** may comprise multiple subsystems. These subsystems may be implemented in software, or hardware, or combinations thereof. As depicted in FIG. **7**, the subsystems may include a user interface subsystem **712** that enables users or customers of cloud infrastructure system **702** to interact with cloud infrastructure system **702**. User interface subsystem **712** may include various different interfaces such as a web interface **714**, an online store interface **716** where cloud services provided by cloud infrastructure system **702** are advertised and are purchasable by a consumer, and other interfaces **718**. For example, a customer may, using a client device, request (service request **734**) one or more services provided by cloud infrastructure system **702** using one or more of interfaces **714**, **716**, and **718**. For example, a customer may access the online store, browse cloud services offered by cloud infrastructure system **702**, and place a subscription order for one or more services offered by cloud infrastructure system **702** that the customer wishes to subscribe to. The service request may include information identifying the customer and one or more services that the customer desires to subscribe to. For example, a customer may place a subscription order for a service offered by cloud infrastructure system **702**. As part of the order, the customer may provide information identifying a chatbot system for which the service is to be provided and optionally one or more credentials for the chatbot system.

In certain examples, such as the example depicted in FIG. **7**, cloud infrastructure system **702** may comprise an order management subsystem (OMS) **720** that is configured to process the new order. As part of this processing, OMS **720** may be configured to: create an account for the customer, if not done already; receive billing and/or accounting information from the customer that is to be used for billing the customer for providing the requested service to the customer; verify the customer information; upon verification, book the order for the customer; and orchestrate various workflows to prepare the order for provisioning.

Once properly validated, OMS **720** may then invoke the order provisioning subsystem (OPS) **724** that is configured to provision resources for the order including processing, memory, and networking resources. The provisioning may include allocating resources for the order and configuring the resources to facilitate the service requested by the customer order. The manner in which resources are provisioned for an order and the type of the provisioned resources may depend upon the type of cloud service that has been ordered by the customer. For example, according to one workflow, OPS **724** may be configured to determine the particular cloud service being requested and identify a number of pods that may have been pre-configured for that particular cloud service. The number of pods that are allocated for an order may depend upon the size/amount/level/scope of the requested service. For example, the number of pods to be allocated may be determined based upon the number of users to be supported by the service, the duration of time for which the service is being requested, and the like. The allocated pods may then be customized for the particular requesting customer for providing the requested service.

In certain examples, setup phase processing, as described above, may be performed by cloud infrastructure system **702** as part of the provisioning process. Cloud infrastructure system **702** may generate an application ID and select a storage virtual machine for an application from among storage virtual machines provided by cloud infrastructure system **702** itself or from storage virtual machines provided by other systems other than cloud infrastructure system **702**.

Cloud infrastructure system **702** may send a response or notification **744** to the requesting customer to indicate when the requested service is now ready for use. In some instances, information (e.g., a link) may be sent to the customer that enables the customer to start using and availing the benefits of the requested services. In certain examples, for a customer requesting the service, the response may include a chatbot system ID generated by cloud infrastructure system **702** and information identifying a chatbot system selected by cloud infrastructure system **702** for the chatbot system corresponding to the chatbot system ID.

Cloud infrastructure system **702** may provide services to multiple customers. For each customer, cloud infrastructure system **702** is responsible for managing information related to one or more subscription orders received from the customer, maintaining customer data related to the orders, and providing the requested services to the customer. Cloud infrastructure system **702** may also collect usage statistics regarding a customer's use of subscribed services. For example, statistics may be collected for the amount of storage used, the amount of data transferred, the number of users, and the amount of system up time and system down time, and the like. This usage information may be used to bill the customer. Billing may be done, for example, on a monthly cycle.

Cloud infrastructure system **702** may provide services to multiple customers in parallel. Cloud infrastructure system **702** may store information for these customers, including possibly proprietary information. In certain examples, cloud infrastructure system **702** comprises an identity management subsystem (IMS) **728** that is configured to manage customer information and provide the separation of the managed information such that information related to one customer is not accessible by another customer. IMS **728** may be configured to provide various security-related services such as identity services, such as information access management, authentication and authorization services, services for managing customer identities and roles and related capabilities, and the like.

FIG. **8** illustrates an example of computer system **800**. In some examples, computer system **800** may be used to implement any of the digital assistant or chatbot systems within a distributed environment, and various servers and computer systems described above. As shown in FIG. **8**, computer system **800** includes various subsystems including a processing subsystem **804** that communicates with a number of other subsystems via a bus subsystem **802**. These other subsystems may include a processing acceleration unit **806**, an I/O subsystem **808**, a storage subsystem **818**, and a communications subsystem **824**. Storage subsystem **818** may include non-transitory computer-readable storage media including storage media **822** and a system memory **810**.

Bus subsystem **802** provides a mechanism for letting the various components and subsystems of computer system **800** communicate with each other as intended. Although bus subsystem **802** is shown schematically as a single bus, alternative examples of the bus subsystem may utilize multiple buses. Bus subsystem **802** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, a local bus using any of a variety of bus architectures, and the like. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which may be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard, and the like.

Processing subsystem **804** controls the operation of computer system **800** and may comprise one or more processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs). The processors may include be single core or multicore processors. The processing resources of computer system **800** may be organized into one or more processing units **832**, **834**, etc. A processing unit may include one or more processors, one or more cores from the same or different processors, a combination of cores and processors, or other combinations of cores and processors. In some examples, processing subsystem **804** may include one or more special purpose co-processors such as graphics processors, digital signal processors (DSPs), or the like. In some examples, some or all of the processing units of processing subsystem **804** may be implemented using customized circuits, such as application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs).

In some examples, the processing units in processing subsystem **804** may execute instructions stored in system memory **810** or on computer readable storage media **822**. In various examples, the processing units may execute a variety of programs or code instructions and may maintain multiple concurrently executing programs or processes. At any given time, some or all of the program code to be executed may be resident in system memory **810** and/or on computer-readable storage media **822** including potentially on one or more storage devices. Through suitable programming, processing subsystem **804** may provide various functionalities described above. In instances where computer system **800** is executing one or more virtual machines, one or more processing units may be allocated to each virtual machine.

In certain examples, a processing acceleration unit **806** may optionally be provided for performing customized processing or for off-loading some of the processing performed by processing subsystem **804** so as to accelerate the overall processing performed by computer system **800**.

I/O subsystem **808** may include devices and mechanisms for inputting information to computer system **800** and/or for outputting information from or via computer system **800**. In general, use of the term input device is intended to include all possible types of devices and mechanisms for inputting information to computer system **800**. User interface input devices may include, for example, a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may also include motion sensing and/or gesture recognition devices such as the Microsoft Kinect® motion sensor that enables users to control and interact with an input device, the Microsoft Xbox® 360 game controller, devices that provide an interface for receiving input using gestures and spoken commands. User interface input devices may also include eye gesture recognition devices such as the Google Glass® blink detector that detects eye activity (e.g., "blinking" while taking pictures and/or making a menu selection) from users and transforms the eye gestures as inputs to an input device (e.g., Google Glass®). Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems (e.g., Siri® navigator) through voice commands.

Other examples of user interface input devices include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, and medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments and the like.

In general, use of the term output device is intended to include all possible types of devices and mechanisms for outputting information from computer system **800** to a user or other computer. User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

Storage subsystem **818** provides a repository or data store for storing information and data that is used by computer system **800**. Storage subsystem **818** provides a tangible non-transitory computer-readable storage medium (e.g., a non-transitory computer-readable memory) for storing the basic programming and data constructs that provide the functionality of some examples. Storage subsystem **818** may store software (e.g., programs, code modules, instructions) that when executed by processing subsystem **804** provides the functionality described above. The software may be executed by one or more processing units of processing subsystem **804**. Storage subsystem **818** may also provide authentication in accordance with the teachings of this disclosure.

Storage subsystem **818** may include one or more non-transitory memory devices, including volatile and non-volatile memory devices. As shown in FIG. **8**, storage subsystem **818** includes a system memory **810** and a computer-readable storage media **822**. System memory **810** may include a number of memories including a volatile main random access memory (RAM) for storage of instructions and data during program execution and a non-volatile read only memory (ROM) or flash memory in which fixed instructions are stored. In some implementations, a basic input/output system (BIOS), containing the basic routines that help to transfer information between elements within computer system **800**, such as during start-up, may typically be stored in the ROM. The RAM typically contains data and/or program modules that are presently being operated and executed by processing subsystem **804**. In some implementations, system memory **810** may include multiple different types of memory, such as static random access memory (SRAM), dynamic random access memory (DRAM), and the like.

By way of example, and not limitation, as depicted in FIG. **8**, system memory **810** may load application programs **812** that are being executed, which may include various applications such as Web browsers, mid-tier applications, relational database management systems (RDBMS), etc., program data **814**, and an operating system **816**. By way of example, operating system **816** may include various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems, a variety of commercially-available UNIX® or UNIX-like operating systems (including without limitation the variety of GNU/Linux operating systems, the Google Chrome® OS, and the like) and/or mobile operating systems such as iOS, Windows® Phone, Android® OS, BlackBerry® OS, Palm® OS operating systems, and others.

Computer-readable storage media **822** may store programming and data constructs that provide the functionality of some examples. Computer-readable media **822** may provide storage of computer-readable instructions, data structures, program modules, and other data for computer system **800**. Software (programs, code modules, instructions) that, when executed by processing subsystem **804** provides the functionality described above, may be stored in storage subsystem **818**. By way of example, computer-readable storage media **822** may include non-volatile memory such as a hard disk drive, a magnetic disk drive, an optical disk drive such as a CD ROM, DVD, a Blu-Ray disk, or other optical media. Computer-readable storage media **822** may include, but is not limited to, Zip® drives, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **822** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magnetoresistive RAM (MRAM) SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs.

In certain examples, storage subsystem **818** may also include a computer-readable storage media reader **820** that may further be connected to computer-readable storage media **822**. Reader **820** may receive and be configured to read data from a memory device such as a disk, a flash drive, etc.

In certain examples, computer system **800** may support virtualization technologies, including but not limited to virtualization of processing and memory resources. For example, computer system **800** may provide support for executing one or more virtual machines. In certain examples, computer system **800** may execute a program such as a hypervisor that facilitated the configuring and managing of the virtual machines. Each virtual machine may be allocated memory, compute (e.g., processors, cores), I/O, and networking resources. Each virtual machine generally runs independently of the other virtual machines. A virtual machine typically runs its own operating system, which may be the same as or different from the operating systems executed by other virtual machines executed by computer system **800**. Accordingly, multiple operating systems may potentially be run concurrently by computer system **800**.

Communications subsystem **824** provides an interface to other computer systems and networks. Communications subsystem **824** serves as an interface for receiving data from and transmitting data to other systems from computer system **800**. For example, communications subsystem **824** may enable computer system **800** to establish a communication channel to one or more client devices via the Internet for receiving and sending information from and to the client devices. For example, when computer system **800** is used to implement bot system **120** depicted in FIG. **1**, the communication subsystem may be used to communicate with a chatbot system selected for an application.

Communication subsystem **824** may support both wired and/or wireless communication protocols. In certain examples, communications subsystem **824** may include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G or EDGE (enhanced data rates for global evolution), WiFi (IEEE 802.XX family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some examples, communications subsystem **824** may provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

Communication subsystem **824** may receive and transmit data in various forms. In some examples, in addition to other forms, communications subsystem **824** may receive input communications in the form of structured and/or unstructured data feeds **826**, event streams **828**, event updates **830**, and the like. For example, communications subsystem **824** may be configured to receive (or send) data feeds **826** in real-time from users of social media networks and/or other communication services such as Twitter® feeds, Facebook® updates, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

In certain examples, communications subsystem **824** may be configured to receive data in the form of continuous data streams, which may include event streams **828** of real-time events and/or event updates **830**, that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g. network monitoring and traffic management applications), clickstream analysis tools, automobile traffic monitoring, and the like.

Communications subsystem **824** may also be configured to communicate data from computer system **800** to other computer systems or networks. The data may be communicated in various different forms such as structured and/or unstructured data feeds **826**, event streams **828**, event updates **830**, and the like to one or more databases that may be in communication with one or more streaming data source computers coupled to computer system **800**.

Computer system **800** may be one of various types, including a handheld portable device (e.g., an iPhone® cellular phone, an iPad® computing tablet, a PDA), a wearable device (e.g., a Google Glass® head mounted display), a personal computer, a workstation, a mainframe, a kiosk, a server rack, or any other data processing system. Due to the ever-changing nature of computers and networks, the description of computer system **800** depicted in FIG. **8** is intended only as a specific example. Many other configurations having more or fewer components than the system depicted in FIG. **8** are possible. Based on the disclosure and teachings provided herein, it should be appreciate there are other ways and/or methods to implement the various examples.

Although specific examples have been described, various modifications, alterations, alternative constructions, and equivalents are possible. Examples are not restricted to operation within certain specific data processing environments, but are free to operate within a plurality of data processing environments. Additionally, although certain examples have been described using a particular series of transactions and steps, it should be apparent to those skilled in the art that this is not intended to be limiting. Although some flowcharts describe operations as a sequential process, many of the operations may be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Various features and aspects of the above-described examples may be used individually or jointly.

Further, while certain examples have been described using a particular combination of hardware and software, it should be recognized that other combinations of hardware and software are also possible. Certain examples may be implemented only in hardware, or only in software, or using combinations thereof. The various processes described herein may be implemented on the same processor or different processors in any combination.

Where devices, systems, components or modules are described as being configured to perform certain operations or functions, such configuration may be accomplished, for example, by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation such as by executing computer instructions or code, or processors or cores programmed to execute code or instructions stored on a non-transitory memory medium, or any combination thereof. Processes may communicate using a variety of techniques including but not limited to conventional techniques for inter-process communications, and different pairs of processes may use different techniques, or the same pair of processes may use different techniques at different times.

Specific details are given in this disclosure to provide a thorough understanding of the examples. However, examples may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the examples. This description provides example examples only, and is not intended to limit the scope, applicability, or configuration of other examples. Rather, the preceding description of the examples will provide those skilled in the art with an enabling description for implementing various examples. Various changes may be made in the function and arrangement of elements.

The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader spirit and scope as set forth in the claims. Thus, although specific examples have been described, these are not intended to be limiting. Various modifications and equivalents are within the scope of the following claims.

In the foregoing specification, aspects of the disclosure are described with reference to specific examples thereof, but those skilled in the art will recognize that the disclosure is not limited thereto. Various features and aspects of the above-described disclosure may be used individually or jointly. Further, examples may be utilized in any number of environments and applications beyond those described herein without departing from the broader spirit and scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive.

In the foregoing description, for the purposes of illustration, methods were described in a particular order. It should be appreciated that in alternate examples, the methods may be performed in a different order than that described. It should also be appreciated that the methods described above may be performed by hardware components or may be embodied in sequences of machine-executable instructions, which may be used to cause a machine, such as a general-purpose or special-purpose processor or logic circuits programmed with the instructions to perform the methods. These machine-executable instructions may be stored on one or more machine readable mediums, such as CD-ROMs or other type of optical disks, floppy diskettes, ROMs, RAMs, EPROMs, EEPROMs, magnetic or optical cards, flash memory, or other types of machine-readable mediums suitable for storing electronic instructions. Alternatively, the methods may be performed by a combination of hardware and software.

Where components are described as being configured to perform certain operations, such configuration may be accomplished, for example, by designing electronic circuits or other hardware to perform the operation, by programming programmable electronic circuits (e.g., microprocessors, or other suitable electronic circuits) to perform the operation, or any combination thereof.

While illustrative examples of the application have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed, and that the appended claims are intended to be construed to include such variations, except as limited by the prior art.

What is claimed is:

1. A computer-implemented method for training a machine learning model to process audio or textual language input, the method comprising:

accessing a machine-learning model, the machine-learning model including one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network;

accessing a training data set;

identifying a dropout parameter, wherein the dropout parameter is a dropout rate;

applying the dropout parameter to the second connection, wherein the second connection is a residual connection, and wherein applying the dropout parameter comprises dropping out the residual connection according to the dropout rate;

training the machine-learning model using the training data set to generate a trained machine-learning model, wherein the second connection is dropped out according to the dropout parameter;

inputting an utterance to the trained machine-learning model to identify an entity; and

based on the identified entity, preparing and providing a response to the utterance.

2. The method of claim **1**, wherein:

the machine learning model includes a plurality of attention heads, each attention head providing an output that is concatenated and multiplied by an attention weight matrix; and

the method further comprising dropping out diagonal entries in the attention weight matrix.

3. The method of claim **1**, wherein the dropout parameter is a hyperparameter of the machine-learning model, the method further comprising:

performing hypertuning to identify the dropout parameter.

4. The method of claim **1**, wherein, when the second connection is dropped out, the machine-learning model is caused to learn contextual information through multi-headed self-attention.

5. The method of claim **4**, wherein:

the training data set comprises one or more utterances, each utterance comprising one or more words that are labeled as an entity and one or more words that are not labeled as entities; and

the contextual information is determined based on the one or more words that are not labeled as entities.

6. The method of claim **1**, wherein providing the response to the utterance comprises providing audio output, via a speaker, responsive to the utterance.

7. The method of claim **1**, wherein providing the response to the utterance comprises providing text output, via a display, responsive to the utterance.

8. A system comprising:

one or more processors; and

a memory coupled to the one or more processors, the memory storing a plurality of instructions executable by the one or more processors, the plurality of instructions comprising instructions that when executed by the one or more processors cause the one or more processors to perform operations comprising:

accessing a machine-learning model, the machine-learning model including one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network;

accessing a training data set;

identifying a dropout parameter, wherein the dropout parameter is a dropout rate;

applying the dropout parameter to the second connection, wherein the second connection is a residual connection, and wherein applying the dropout parameter comprises dropping out the residual connection according to the dropout rate;

training the machine-learning model using the training data set to generate a trained machine-learning model, wherein the second connection is dropped out according to the dropout parameter;

inputting an utterance to the trained machine-learning model to identify an entity; and

based on the identified entity, preparing and providing a response to the utterance.

9. The system of claim **8**, wherein:

the machine learning model includes a plurality of attention heads, each attention head providing an output that is concatenated and multiplied by an attention weight matrix; and

the operations further comprise dropping out diagonal entries in the attention weight matrix.

10. The system of claim **8**, wherein:

the dropout parameter is a hyperparameter of the machine-learning model, and

the operations further comprise performing hypertuning to identify the dropout parameter.

11. The system of claim **8**, wherein:

when the second connection is dropped out, the machine-learning model is caused to learn contextual information through multi-headed self-attention.

12. The system of claim **11**, wherein:

the training data set comprises one or more utterances, each utterance comprising one or more words that are labeled as an entity and one or more words that are not labeled as entities; and

the contextual information is determined based on the one or more words that are not labeled as entities.

13. The system of claim **11**, wherein providing the response to the utterance comprises providing audio output, via a speaker, responsive to the utterance.

14. The system of claim **11**, wherein providing the response to the utterance comprises providing text output, via a display, responsive to the utterance.

15. A non-transitory computer-readable memory storing a plurality of instructions executable by one or more processors to cause the one or more processors to perform operations comprising:

accessing a machine-learning model, the machine-learning model including one or more blocks in which each block includes a multi-head self-attention network, a first connection for providing input to the multi-head self-attention network, and a second connection for providing the input to a normalization layer, bypassing the multi-head self-attention network;

accessing a training data set;

identifying a dropout parameter, wherein the dropout parameter is a dropout rate;

applying the dropout parameter to the second connection, wherein the second connection is a residual connection, and wherein applying the dropout parameter comprises dropping out the residual connection according to the dropout rate;

training the machine-learning model using the training data set to generate a trained machine-learning model, wherein the second connection is dropped out according to the dropout parameter;

inputting an utterance to the trained machine-learning model to identify an entity; and

based on the identified entity, preparing and providing a response to the utterance.

16. The non-transitory computer-readable memory of claim **15**, wherein:

the machine learning model includes a plurality of attention heads, each attention head providing an output that is concatenated and multiplied by an attention weight matrix; and

the operations further comprise dropping out diagonal entries in the attention weight matrix.

17. The non-transitory computer-readable memory of claim **15**, wherein:

the dropout parameter is a hyperparameter of the machine-learning model, and

the operations further comprise performing hypertuning to identify the dropout parameter.

18. The non-transitory computer-readable memory of claim **15**, wherein:

when the second connection is dropped out, the machine-learning model is caused to learn contextual information through multi-headed self-attention.

19. The non-transitory computer-readable memory of claim **15**, wherein providing the response to the utterance comprises providing audio output, via a speaker, responsive to the utterance.

20. The non-transitory computer-readable memory of claim **15**, wherein providing the response to the utterance comprises providing text output, via a display, responsive to the utterance.

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